

Brzozowski Goes Concurrent

A Kleene Theorem for Pomset Languages

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Abstract

Concurrent Kleene Algebra (CKA) is a mathematical formalism to study programs that exhibit concurrent behaviour. As with previous extensions of Kleene Algebra, characterizing the free model is crucial in order to develop the foundations of the theory and potential applications. For CKA, this has been an open question for a few years and this paper makes an important step towards an answer. We present a new automaton model and a Kleene-like theorem for pomset languages, which are a natural candidate for the free model. There are two substantial differences with previous work: from expressions to automata, we use Brzozowski derivatives, which enable a direct construction of the automaton; from automata to expressions, we provide a syntactic characterization of the automata that denote valid CKA behaviours.

1 Introduction

In their CONCUR'09 paper [4], Hoare, Möller, Struth, and Wehrman introduced Concurrent Kleene Algebra (CKA) as a suitable mathematical framework to study concurrent programs, in the hope of achieving the same elegance that Kozen did when using Kleene Algebra (and extensions) to provide a verification platform for sequential programs.

CKA is a seemingly simple extension of Kleene Algebra (KA): it adds a parallel operator that allows to compositionally specify concurrent behaviours. However, extending the existing KA toolkit — importantly, completeness and decidability results — turns out to be challenging. A fundamental missing ingredient is a characterization of the free model for CKA. This is in striking contrast with KA, where these topics are well understood. Several authors [5, 7] have conjectured the free model to be pomset languages — a generalization of regular languages to sets of partially ordered words.

In KA, Kleene's theorem provided a pillar for developing the toolkit and axiomatization [11], and, by extension, characterizing the free model. In this light, we pursue a Kleene Theorem for CKA. Specifically, we study CKA expressions, also called series-rational expressions, with a denotational model given in terms of pomset languages. Our main contribution is a Kleene Theorem for series-rational expressions, based on two constructions faithfully translating between the denotational model and a newly defined operational model, which we call *pomset automata*. In a nutshell, these are finite-state automata in which computations from a certain state s may branch into parallel threads that contribute to the language of s whenever they both reach a final state.

We are not the first to attempt such a Kleene theorem. However, earlier works [14, 7] fall short of giving a precise correspondence between the denotational and operational models, due to the lack of a suitable automata restriction ensuring that only valid behaviours are accepted. We overcome this situation by introducing a generalization of Brzozowski derivatives [2] in the translation from expressions to automata. This guides us to a *syntactic* restriction on automata (rather than the *semantic* condition put forward in previous works), which guarantees the existence of a reverse construction, from automata to expressions. Moreover, following the Brzozowski route allows us to bypass a Thompson-like construction [16], avoiding the introduction of ϵ -transitions and non-determinism present in the aforementioned works.

The remainder of this paper is organized as follows. In Section 2, we introduce the necessary notation. In Section 3, we introduce our automaton model as well as some notable subclasses of automata. In

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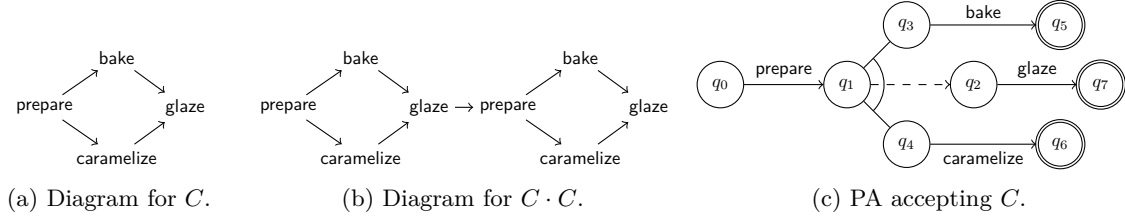


Figure 1: Hasse diagrams for pomsets and a pomset automaton accepting one.

Section 4, we discuss how to translate a series-rational expression to a semantically equivalent pomset automaton, while in Section 5 we show how to translate a suitably restricted class of pomset automata to series-rational expressions. We contrast results with earlier work in Section 6. Directions for further work in are listed in Section 7.

To preserve the flow of the narrative, proofs of the more routine lemmas appear in Appendix A.

2 Preliminaries

For S a set, we write 2^S for the set of all subsets of S , and $\binom{S}{2}$ for the set of *multisets* over S of size two. An element of $\binom{S}{2}$ containing $s_1, s_2 \in S$ is written $\{s_1, s_2\}$; note that $\{s_1, s_2\} = \{s_2, s_1\}$. We use the symbols ϕ and ψ to denote multisets. If S and I are sets, and for every $i \in I$ there exists an $s_i \in S$, we call $(s_i)_{i \in I}$ an *I -indexed family over S* . We sometimes write $(s_i)_{i \in I} \in S$ to signify that the latter is a family over S . We say that a relation $\prec \subseteq S \times S$ is a *strict order* on S if it is irreflexive, asymmetric and transitive. We refer to $\prec$ as *well-founded* if there are no infinite descending $\prec$ -chains, i.e., no family $(s_n)_{n \in \mathbb{N}} \in S$ such that $\forall n \in \mathbb{N}, s_{n+1} \prec s_n$. Throughout the paper we fix a finite set Σ called the *alphabet*, whose elements are symbols usually denoted with a and b . Lastly, if $\rightarrow \subseteq X \times Y \times Z$ is a ternary relation, we write $x \xrightarrow{y} z$ instead of $\langle x, y, z \rangle \in \rightarrow$.

2.1 Pomsets

Partially-ordered multiset, or *pomset* [3] for short, generalise words to a setting where events (elements from Σ) may take place not just sequentially, but also in parallel.

Definition 2.1. A *labelled poset* is a tuple $\langle U, \leq_U, \lambda_U \rangle$ consisting of a *carrier set* U , a partial order $\leq_U$ on U and a *labelling function* $\lambda_U : U \rightarrow \Sigma$. A *labelled poset isomorphism* is a bijection between poset carriers that bijectively preserves the labels and the ordering. A *pomset* is an isomorphism class of labelled posets; in practice, it is a labelled poset up-to bijective renaming of elements in U . We write 1 for the empty pomset, Pom_Σ for the set of all pomsets and Pom_Σ^+ for the set of all the non-empty pomsets.

For instance, suppose a recipe for caramel-glazed cookies tells us to (i) *prepare* cookie dough (ii) *bake* cookies in the oven (iii) *caramelize* sugar (iv) *glaze* the finished cookies. Here, step (i) precedes steps (ii) and (iii). Furthermore, step (iv) succeeds both steps (ii) and (iii). A pomset representing this process could be $\langle C, \leq_C, \lambda_C \rangle$, where $C = \{(i), (ii), (iii), (iv)\}$ and $\leq_C$ is such that (i) $\leq_C$ (ii) $\leq_C$ (iv) and (i) $\leq_C$ (iii) $\leq_C$ (iv); λ_C is as in the recipe.

Note that words are just finite pomsets with a total order. We will sometimes refer to $a \in \Sigma$ as the pomset with a single point labelled a (and the obvious order); such a pomset is called *primitive*. A pomset can be represented as a Hasse diagram, where nodes have labels in Σ . For instance, the Hasse diagram for the pomset C above is drawn in Figure 1a.

To simplify notation, we refer to a pomset by the carrier U of a labelled poset $\langle U, \leq_U, \lambda_U \rangle$ in this isomorphism class. We use the symbols U, V, W and X to denote pomsets. Pomsets being isomorphism classes, the content of the carrier of the chosen representative is of very little importance; it is the order and labelling that matters. For this reason, we tacitly assume that whenever we have two pomsets, we pick representatives that have disjoint carrier sets.

Definition 2.2. The *width* of a pomset U , denoted $\|U\|$, is the size of the largest antichain in U with respect to $\leq_U$, i.e., the maximum $n \in \mathbb{N}$ such that there exist $u_1, u_2, \dots, u_n \in U$ that are not related by $\leq_U$.

The pomsets we work with in this paper have a finite carrier. As a result, $\|U\|$ is always defined. For instance, the width of the pomset C above is 2, because the nodes (ii) and (iii) are an antichain of size 2, and there is no antichain of size 3.

Definition 2.3. Let U and V be pomsets. The *sequential composition* of U and V , denoted $U \cdot V$, is the pomset $\langle U \cup V, \leq_U \cup \leq_V \cup (U \times V), \lambda_U \cup \lambda_V \rangle$. The *parallel composition* of U and V , denoted $U \parallel V$, is the pomset $\langle U \cup V, \leq_U \cup \leq_V, \lambda_U \cup \lambda_V \rangle$. Here, $\lambda_U \cup \lambda_V$ is the function from $U \cup V$ to Σ that agrees with λ_U on U , and with λ_V on V .

Note that 1 is the unit for both sequential and parallel composition. Sequential composition forces the events in the left pomset to be ordered before those in the right pomset. An example, describing the pomset $C \cdot C$, is depicted in Figure 1b.

Definition 2.4. The set of *series-rational* pomsets is the smallest set that includes the primitive pomsets and is closed under sequential and parallel composition.

In this paper we will be mostly concerned with series-rational pomsets. For inductive reasoning about them, it is useful to record the following lemma.

Lemma 2.5 (Gischer [3]). *Let U be a series-rational pomset. Then, either (i) $U = a$ for some $a \in \Sigma$, or (ii) $U = V \cdot W$ for series-rational $V, W \in \text{Pom}_{\Sigma}^+$, both strictly smaller than U , or (iii) $U = V \parallel W$ for series-rational $V, W \in \text{Pom}_{\Sigma}^+$, both strictly smaller than U .*

2.2 Pomset languages

If a sequential program can exhibit multiple traces, we can group the words that represent these traces into a set called a *language*. By analogy, we can group the pomsets that represent the traces that arise from a parallel program into a set, which we refer to as a *pomset language*. Pomset languages are denoted by the symbols $\mathcal{U}$ and $\mathcal{V}$.

For instance, suppose that the recipe for glazed cookies may have an optional fifth step where chocolate sprinkles are spread over the cookies. In that case, there are *two* pomsets that describe a trace arising from the recipe, C^+ and C^- , either with or without the chocolate sprinkles. The pomset language $\mathcal{C} = \{C^-, C^+\}$ describes the new recipe.

Definition 2.6. Let $\mathcal{U}$ be a pomset language. $\mathcal{U}$ has *bounded width* if there is $n \in \mathbb{N}$ such that for all $U \in \mathcal{U}$ we have $\|U\| \leq n$. The minimal such n is the *width* of $\mathcal{U}$, written $\|\mathcal{U}\|$.

The pomset languages considered in this paper have bounded width, and hence $\|\mathcal{U}\|$ is always defined. For instance, the width of $\mathcal{C}$ is 2, because the width of both C^+ and C^- is 2.

The sequential and parallel compositions of pomsets can be lifted to pomset languages. We also define a Kleene closure operator, similar to the one defined on languages of words.

Definition 2.7. Let $\mathcal{U}$ and $\mathcal{V}$ be pomset languages. We define:

$$\mathcal{U} \cdot \mathcal{V} = \{U \cdot V : U \in \mathcal{U}, V \in \mathcal{V}\} \quad \mathcal{U} \parallel \mathcal{V} = \{U \parallel V : U \in \mathcal{U}, V \in \mathcal{V}\} \quad \mathcal{U}^* = \bigcup_{n \in \mathbb{N}} \mathcal{U}^n$$

Where $\mathcal{U}^0 = \{1\}$, and $\mathcal{U}^{n+1} = \mathcal{U} \cdot \mathcal{U}^n$ for all $n \in \mathbb{N}$.

Kleene closure models indefinite repetition. For instance, if our cookie recipe has a final step “repeat until enough cookies have been made”, the pomset language $\mathcal{C}^*$ represents all possible traces of repetitions of the recipe; e.g., $C^+ \cdot C^+ \cdot C^- \in \mathcal{C}^*$ is the trace where first two batches of sprinkled cookies are made, followed by one without sprinkles.

2.3 Series-rational expressions

Just like a rational expression can be used to describe a regular structure of sequential events, a series-rational expression can be used to describe a regular structure of possibly parallel events. Series-rational expressions are rational expressions with parallel composition.

Definition 2.8. The *series-rational expressions*, denoted $\mathcal{T}_{\Sigma}$, are formed by the grammar

$$e, f ::= 0 \mid 1 \mid a \in \Sigma \mid e + f \mid e \cdot f \mid e \parallel f \mid e^*$$

We use the symbols d, e, f, g and h to denote series-rational expressions.

The semantics of a series-rational expression is given by a pomset language.

Definition 2.9. The function $\llbracket - \rrbracket : \mathcal{T}_\Sigma \rightarrow 2^{\text{Pom}_\Sigma}$ is defined inductively, as follows:

$$\begin{aligned}\llbracket 0 \rrbracket &= \emptyset \\ \llbracket a \rrbracket &= \{a\} \\ \llbracket e \cdot f \rrbracket &= \llbracket e \rrbracket \cdot \llbracket f \rrbracket \\ \llbracket e^* \rrbracket &= \llbracket e \rrbracket^* \\ \llbracket 1 \rrbracket &= \{1\} \\ \llbracket e + f \rrbracket &= \llbracket e \rrbracket \cup \llbracket f \rrbracket \\ \llbracket e \parallel f \rrbracket &= \llbracket e \rrbracket \parallel \llbracket f \rrbracket\end{aligned}$$

If $\mathcal{U} \in 2^{\text{Pom}_\Sigma}$ such that $\mathcal{U} = \llbracket e \rrbracket$ for some $e \in \mathcal{T}_\Sigma$, then $\mathcal{U}$ is a *series-rational language*.

To illustrate, consider the pomset language $\mathcal{C}^* = \{C^+, C^-\}^*$, which describes the possible traces arising from indefinitely repeating the cookie recipe, optionally adding chocolate sprinkles at every repetition. We can describe the pomset language $\{C^-\}$ with the series-rational expression $c^- = \text{prepare} \cdot (\text{bake} \parallel \text{caramelize}) \cdot \text{glaze}$, and $\{C^+\}$ by $c^+ = c^- \cdot \text{sprinkle}$, which yields the series-rational expression $c = c^- + c^+$ for $\mathcal{C}$. By construction, $\llbracket c^* \rrbracket = \mathcal{C}^*$.

2.4 Additive congruence

From the definition of $\llbracket - \rrbracket$, we can already see that series-rational expressions that are equal modulo associativity, commutativity and idempotence of the $+$ -operator, as well as the unit 0, are also semantically equivalent. We furthermore identify series-rational expressions with an empty language. We formalize this identification by defining an congruence relation on series-rational expressions that will be useful throughout the paper.

Definition 2.10. We define $\simeq$ as the smallest congruence on $\mathcal{T}_\Sigma$ such that:

$$\begin{aligned}e_1 + 0 &\simeq e_1 & e_1 + e_1 &\simeq e_1 & e_1 + e_2 &\simeq e_2 + e_1 & e_1 + (e_2 + e_3) &\simeq (e_1 + e_2) + e_3 \\ 0 \cdot e_1 &\simeq 0 & e_1 \cdot 0 &\simeq 0 & 0 \parallel e &\simeq 0 & e \parallel 0 &\simeq 0\end{aligned}$$

When $\{g, h\}, \{g', h'\} \in \binom{\mathcal{T}_\Sigma}{2}$ such that $g \simeq g'$ and $h \simeq h'$, we write $\{g, h\} \simeq \{g', h'\}$.

Thus, when we claim that $e \simeq e'$, we say that e is equal to e' , modulo the removal or addition of 0-terms, the merging of identical terms and the reordering or re-bracketing of terms (possibly within subterms). Moreover, congruence is sound with respect to the semantics, and it identifies all expressions that have an empty denotational semantics.

Lemma 2.11. Let $e, f \in \mathcal{T}_\Sigma$. If $e \simeq f$, then $\llbracket e \rrbracket = \llbracket f \rrbracket$. Also, $e \simeq 0$ if and only if $\llbracket e \rrbracket = \emptyset$.

Proof. Refer to Appendix A. □

There is a simple linear time decision procedure to test whether two expressions are congruent. This justifies our using this relation to build finite automata later on. As a by-product, we get that the emptiness problem for series-rational expressions is linear time decidable.

3 Pomset Automata

We are now ready to describe an automaton model that recognises series-rational languages.

Definition 3.1. A *pomset automaton (PA)* is a tuple $\langle Q, \delta, \gamma, F \rangle$ where Q is a set of *states*, with $F \subseteq Q$ the *accepting states*, $\delta : Q \times \Sigma \rightarrow Q$ is a function called the *sequential transition function*, $\gamma : Q \times \binom{Q}{2} \rightarrow Q$ is a function called the *parallel transition function*.

Note that we do not fix an initial state. As a result, a PA does not define a single pomset language but rather a mapping from its states to pomset languages. The language of a state is defined in terms of a trace relation that involves the transitions of both δ and γ . For the remainder of this section, we fix a PA $A = \langle Q, \delta, \gamma, F \rangle$, and a state $q \in Q$. To simplify matters later on, we assume that A has a state

$\perp \in Q - F$ such that, for every $a \in \Sigma$, it holds that $\delta(\perp, a) = \perp$ and, for every $\phi \in \binom{Q}{2}$, it holds that $\gamma(\perp, \phi) = \perp$. Such a *sink state* is particularly useful for γ : for a fixed $q \in Q$ not all $\phi \in \binom{Q}{2}$ may give a value of $\gamma(q, \phi)$ that contributes to the language accepted by q . These inputs can be mapped to $\perp$ by γ . Equivalently, we can allow γ to be a partial function; we chose γ as a total function so as not to clutter the definition of derivatives in Section 4.

Definition 3.2. $\rightarrow_A \subseteq Q \times \text{Pom}_\Sigma^+ \times Q$ is the smallest relation satisfying the rules

$$\frac{}{q \xrightarrow{a}_A \delta(q, a)} \quad \frac{q \xrightarrow{U}_A q'' \quad q'' \xrightarrow{V}_A q'}{q \xrightarrow{U \cdot V}_A q'} \quad \frac{r \xrightarrow{U}_A r' \in F \quad s \xrightarrow{V}_A s' \in F}{q \xrightarrow{U \parallel V}_A \gamma(q, \{r, s\})}$$

We also define $\twoheadrightarrow_A \subseteq Q \times \text{Pom}_\Sigma \times Q$ by $q \twoheadrightarrow_A q'$ if and only if $q' = q$ and $U = 1$, or $q \xrightarrow{U}_A q'$. The *language* of A at $q \in Q$, denoted $L_A(q)$, is the set $\{U : \exists q' \in F. q \twoheadrightarrow_A q'\}$. We say that A *accepts* the language $\mathcal{U}$ if there exists a $q \in Q$ such that $L_A(q) = \mathcal{U}$.

Intuitively, γ ensures that when a process forks at state q into subprocesses starting at r and s , if each of those reaches an accepting state, then the processes can join at $\gamma(q, \{r, s\})$.

We purposefully omit the empty pomset 1 as a label in $\rightarrow_A$; doing so would open up the possibility of having traces of the form $q \xrightarrow{1}_A q'$ with $q \neq q'$ (i.e., “silent transitions” or “ ϵ -transitions”) for example by defining $\gamma(q, \{r, s\}) = q'$ for some $r, s \in F$. Avoiding transitions of this kind allows us to prove claims about $\rightarrow_A$ by induction on the pomset size, and leverage Lemma 2.5 in the process to disambiguate between the rules that apply. By extension, we can prove claims about $\twoheadrightarrow_A$ and L_A by treating $U = 1$ as a special case.

We draw a PA in a way similar to finite automata: each state (except $\perp$) is a vertex, and accepting states are marked by a double border. To represent sequential transitions, we draw labelled edges; for instance, in Figure 1c, $\delta(q_0, \text{prepare}) = q_1$. To represent parallel transitions, we draw hyper-edges; for instance, in Figure 1c, $\gamma(q_1, \{q_3, q_4\}) = q_2$. To avoid clutter, we do not draw either of these edges types the target state is $\perp$. It is not hard to verify that the pomset C of the earlier example is accepted by the PA in Figure 1c.

In principle, the state space of a PA can be infinite; we use this later on to define a PA that has all possible series-rational expressions as states. It is however also useful to know when we can prune an infinite PA into a finite PA while preserving some accepted languages.

Note that it is not sufficient to talk about reachable states, i.e., states that appear in the target of some trace; we must also include states that are “meaningful” starting points for subprocesses. To do this, we first need a handle on these starting points. Specifically, we are interested in the states where (1) joining and forking yields a state that contributes to the behaviour of the PA, and (2) these states may join again, because they are not the sink state. This is captured in the definition below.

Definition 3.3. The *support* of q , written $\pi_A(q)$, is the largest subset of $\binom{Q}{2}$ such that if $\{r, s\} \in \pi_A(q)$, then $\gamma(q, \{r, s\}) \neq \perp$, and $r, s \neq \perp$.

We can now talk about subsets of states of an automaton that are closed, in the sense that the relevant part of a transition function has input and output confined to this set. As a result, we can confine the structure of a given PA to a closed set.

Definition 3.4. A set of states $Q' \subseteq Q$ is *closed* when the following rules are satisfied

$$\frac{}{\perp \in Q'} \quad \frac{q \in Q' \quad a \in \Sigma}{\delta(q, a) \in Q'} \quad \frac{q \in Q' \quad \phi \in \pi_A(q)}{\gamma(q, \phi) \in Q'} \quad \frac{q \in Q' \quad \{r, s\} \in \pi_A(q)}{r, s \in Q'}$$

If Q' is closed, the *generated sub-PA* of A induced by Q' , denoted $A \upharpoonright_{Q'}$, is the tuple $\langle Q', \delta \upharpoonright_{Q'}, \gamma \upharpoonright_{Q'}, Q' \cap F \rangle$ where $\delta \upharpoonright_{Q'}$ and $\gamma \upharpoonright_{Q'}$ are the restrictions of δ and γ to Q' .

Because the relevant parts of the transition functions are preserved, it is not altogether surprising that languages of a generated sub-PA are also languages of the original PA.

Lemma 3.5. Let $Q' \subseteq Q$ be closed. If $q \in Q'$, then $L_{A \upharpoonright_{Q'}}(q) = L_A(q)$.

Proof. First, observe that $1 \in L_{A \upharpoonright_{Q'}}(q)$ if and only if $1 \in L_A(q)$, since $q \in Q' \cap F$ if and only if $q \in F$. Now, suppose that $q \xrightarrow{U}_{A \upharpoonright_{Q'}} q'$; we show that in this case $q \xrightarrow{U}_A q'$ follows. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$, and $q' = \delta \upharpoonright_{Q'}(q, a)$. But then $q' = \delta(q, a)$, and so $q \xrightarrow{U}_A q'$. For the inductive step, there are two cases to consider:

- If $U = V \cdot W$ with V and W smaller than U , there exists a $q'' \in Q'$ such that $q \xrightarrow{V}_{A'} q''$ and $q'' \xrightarrow{W}_{A'} q'$. By induction, we find $q \xrightarrow{V}_A q''$ and $q'' \xrightarrow{W}_A q'$, and thus $q \xrightarrow{U}_A q'$.
- If $U = V \parallel W$ with V and W smaller than U , then there exist $r, s \in Q'$ and $r', s' \in F \cap Q'$ such that $r \xrightarrow{V}_{A'} r'$ and $s \xrightarrow{W}_{A'} s'$, and $q' = \gamma \upharpoonright_{Q'} (q, \llbracket r, s \rrbracket)$. By induction we find that $r \xrightarrow{V}_A r'$ and $s \xrightarrow{W}_A s'$; note that $r', s' \in F$. It then follows that $q \xrightarrow{U}_A \gamma(q, \llbracket r, s \rrbracket) = q'$.

For the other direction, suppose that $q \xrightarrow{U}_A q'$ and $q \neq \perp$; we show that in this case $q \xrightarrow{U}_{A'} q'$ follows. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$ and $q' = \delta(q, a)$. But then $q' = \delta \upharpoonright_{Q'} (q, a)$ and so $q \xrightarrow{U}_{A'} q'$. For the inductive step, there are two cases to consider:

- If $U = V \cdot W$ with V and W smaller than U , there exists a $q'' \in Q$ such that $q \xrightarrow{V}_{A'} q''$ and $q'' \xrightarrow{W}_{A'} q'$. Since $q' \neq \perp$, we have that $q'' \neq \perp$. By induction we then find that $q \xrightarrow{V}_A q''$ (thus $q'' \in Q'$) and so again by induction $q'' \xrightarrow{W}_{A'} q'$. In total, we obtain $q \xrightarrow{U}_{A'} q'$.
- If $U = V \parallel W$ with V and W smaller than U , there exist $r, s \in Q$ and $r', s' \in F$ such that $r \xrightarrow{V}_{A'} r'$ and $s \xrightarrow{W}_{A'} s'$, and $q' = \gamma(q, \llbracket r, s \rrbracket)$. Since $q' \neq \perp$, it holds that $\llbracket r, s \rrbracket \in \pi_A(q)$, thus $r, s \in Q'$; also, since $r', s' \in F$, we know that $r', s' \neq \perp$. By induction we find that $r \xrightarrow{V}_{A'} r'$ and $s \xrightarrow{W}_{A'} s'$. Note that, since $r', s' \in F$, also $r, s \neq \perp$, and therefore $\llbracket r, s \rrbracket \in \pi_A(q)$. We then conclude that $q \xrightarrow{U}_{A'} \gamma \upharpoonright_{Q'} (q, \llbracket r, s \rrbracket) = \gamma(q, \llbracket r, s \rrbracket) = q'$. $\square$

We now work out how to find a closed subset of states that contains a particular state. The first step is to characterize the states reachable from q by means of a trace.

Definition 3.6. The *reach* of q , written $\rho_A(q)$, is the smallest set satisfying the rules

$$\frac{}{q, \perp \in \rho_A(q)} \quad \frac{q' \in \rho_A(q) \quad a \in \Sigma}{\delta(q', a) \in \rho_A(q)} \quad \frac{q' \in \rho_A(q) \quad \phi \in \pi_A(q)}{\gamma(q', \phi) \in \rho_A(q)}$$

The reach of a state is closely connected to the states that can be reached from q through the trace relation of the automaton, in the following way:

Lemma 3.7. *The reach of q contains $\{q' \in Q : \exists U \in \text{Pom}_\Sigma^+. q \xrightarrow{U}_A q'\} \cup \{\perp, q\}$. Moreover, if $\perp$ is the only state of A whose language is empty, this containment is an equality.*

Proof. Refer to Appendix A. $\square$

Note that $\rho_A(q)$ is not necessarily closed: we also need the states required by the fourth rule of closure in Definition 3.4. Thus, if we want to “close” $\rho_A(q)$ by adding the support of its contents, we need to find finite, closed sets of states that contain branching points. In order to do this inductively, we propose the following subclass of PAs.

Definition 3.8. We say that A is *fork-acyclic* if there exists a *fork hierarchy*, which is a strict order $\prec_A \subseteq Q \times Q$ such that the following rules are satisfied.

$$\frac{\llbracket r, s \rrbracket \in \pi_A(q)}{r, s \prec_A q} \quad \frac{a \in \Sigma \quad r \prec_A \delta(q, a)}{r \prec_A q} \quad \frac{\phi \in \pi_A(q) \quad r \prec_A \gamma(q, \phi)}{r \prec_A q}$$

The fork hierarchy is connected with the reach of a state in the following way.

Lemma 3.9. *Let $q', r \in Q$. If A is fork-acyclic, $q' \in \rho_A(q)$ and $r \prec_A q'$, then $r \prec_A q$.*

Proof. The proof proceeds by induction on the construction of $\rho_A(q)$. In the base, either $q' = q$ or $q' = \perp$. In the former case, the claim holds vacuously.

The latter case can be excluded, since none of the rules for a fork hierarchy mandate that $r \prec_A \perp$ should hold: the first rule does not apply since $\pi_A(\perp)$ is empty, and the second and third rules do not apply because $\delta(\perp, a) = \perp$ and $\gamma(\perp, \phi) = \perp$.

For the inductive step, there are two cases to consider.

- If $q' = \delta(q'', a)$ for some $q'' \in \rho_A(q)$ and $a \in \Sigma$, then we know that $r \prec_A q''$ by definition of $\prec_A$. But then $r \prec_A q$ by induction.
- If $q' = \gamma(q'', \phi)$ for some $q'' \in \rho_A(q)$ and $\phi \in \pi_A(q'')$, then we know that $r \prec_A q''$ by definition of $\prec_A$. But then $r \prec_A q$ by induction. $\square$

The term *fork-acyclic* has been used in literature for similar automata [14, 6]. However, in op. cit., it is defined in terms of the traces that arise from the transition structure of the automaton. In contrast, our definition is purely syntactic: it imposes an order on states such that forks cannot be nested. To show that, as in [14], our definition similarly implies languages of the PA have bounded width, we present the following lemma. Since the state space of a PA can be infinite, we additionally require that the fork hierarchy is well-founded.

Lemma 3.10. *If A is fork-acyclic and $\prec_A$ is well-founded then $L_A(q)$ is of finite width.*

Proof. If $q \in Q$, let $n_q \in \mathbb{N}$ be the lowest upper bound on the length of a descending chain starting at q , i.e., if $r_1, r_2, \dots, r_m \in Q$ are such that $r_1 \prec_A r_2 \prec_A \dots \prec_A r_m \prec_A q$, then $m \leq n_q$. Such an n_q exists uniquely, for $\prec_A$ is well-founded. We strengthen our claim as follows: if $q \in Q$, then $\|L_A(q)\| \leq 2^{n_q}$.

It suffices to show that if $q' \in Q$ and $U \in \mathbf{Pom}_\Sigma^+$ are such that $q \xrightarrow{U}_A q'$ and $q' \neq \perp$, then $\|U\| \leq 2^{n_q}$. The proof proceeds by nested induction. The outer induction is on n_q ; here, we assume the claim holds for all $r \in Q$ with $n_r < n_q$ (note that this implicitly covers the base, where $n_q = 0$). The inner induction is on U . In the base of the inner induction, $U = a \in \Sigma$ and the claim holds immediately, for $\|a\| = 1 \leq 2^{n_q}$.

For the inductive step of the inner induction, there are two cases to consider.

- If $U = V \cdot W$, with V and W smaller than U , then there exists a $q'' \in Q$ such that $q \xrightarrow{V}_A q''$ and $q'' \xrightarrow{W}_A q'$. Since $q' \neq \perp$, we know that $q'' \neq \perp$ as well. Therefore, we know by the (inner and outer) induction hypothesis that $\|V\| \leq 2^{n_q}$. Furthermore, since $q'' \in \rho_A(q)$ by Lemma 3.7, we know that $q'' \prec_A q$ by Lemma 3.9, therefore $n_{q''} < n_q$. Thus, by the (inner) induction hypothesis, it follows that $\|W\| \leq 2^{n_{q''}}$. But then $\|U\| = \max(\|V\|, \|W\|) \leq \max(2^{n_q}, 2^{n_{q''}}) = 2^{n_q}$ as well.
- If $U = V \parallel W$, with V and W smaller than U , then there exist $r, s \in Q$ and $r', s' \in F$ such that $r \xrightarrow{V}_A r'$ and $s \xrightarrow{W}_A s'$. By the outer induction hypothesis, we know that $\|V\| \leq \|L_A(r)\| \leq 2^{n_r}$ and $\|W\| \leq \|L_A(s)\| \leq 2^{n_s}$. Furthermore, since $q', r, s \neq \perp$, we know that $\{r, s\} \in \pi_A(q)$, thus $r, s \prec_A q$ and therefore $n_s, n_r < n_q$. We then derive

$$\|U\| = \|V\| + \|W\| \leq 2^{n_r} + 2^{n_s} \leq 2 \cdot 2^{\max(n_r, n_s)} = 2^{\max(n_r, n_s) + 1} \leq 2^{n_q}$$

This completes the proof. $\square$

We introduce the notion of a *bounded PA*, which is sufficient to guarantee the existence of a closed, finite subset containing a given state, even when the PA has infinitely many states.

Definition 3.11. If A is fork-acyclic, we say that it is *bounded* if $\prec_A$ is well-founded, and for all $q \in Q$, both $\pi_A(q)$ and $\rho_A(q)$ are finite.

Theorem 3.12. *If A is bounded, there exists a finite set of states $Q_q \subseteq Q$ that is closed and contains q .*

Proof. The proof proceeds by $\prec_A$ -induction; this is sound, because $\prec_A$ is well-founded.

Suppose the claim holds for all $r \in Q$ with $r \prec_A q$. If $q' \in \rho_A(q)$ and $\{r, s\} \in \pi_A(q')$, then $r \prec_A q'$ and thus $r \prec_A q$ by Lemma 3.9; by induction we obtain for every such r a finite set of states $Q_r \subseteq Q$ that is closed in A and contains r . We choose:

$$Q_q = \rho_A(q) \cup \bigcup \{Q_r : q' \in \rho_A(q), \{r, s\} \in \pi_A(q')\}$$

This set is finite because $\rho_A(q)$ and $\pi_A(q')$ are finite for all $q, q' \in Q$ since A is bounded. To see that Q_q is closed in A , it suffices to show that the last rule of closure holds for $q' \in \rho_A(q)$; it does, since if $q' \in \rho_A(q)$ and $\{r, s\} \in \pi_A(q')$, then $r \in Q_r$ and $s \in Q_s$, thus $r, s \in Q_q$. $\square$

3.1 Lock-step traces

Note that while the transition functions of a PA are deterministic in that their output is a single state rather than a set of states, the transition relation of a PA $A = \langle Q, \delta, \gamma, F \rangle$ should not be thought of as deterministic. In particular, if $q \xrightarrow{U}_A q'$ and $q \xrightarrow{U}_A q''$, then $q' = q''$ does not hold in general. For instance, consider the case where $\gamma(q, \{r_1, s_1\}) = q'$ and $\gamma(q, \{r_2, s_2\}) = q''$ and furthermore $s_1 \xrightarrow{V}_A s'_1 \in F$ and $s_2 \xrightarrow{V}_A s'_2 \in F$ as well as $r_1 \xrightarrow{W}_A r'_1 \in F$ and $r_2 \xrightarrow{W}_A r'_2 \in F$. Here, $q \xrightarrow{V \parallel W}_A q'$ and $q \xrightarrow{V \parallel W}_A q''$, but q' and q'' may not be equal.

When working with traces in a PA, it may be useful to be able to relate traces of the same pomset in the presence of this kind of determinism. To this end, we introduce the notion of lock-step traces. Intuitively, lock-step traces prevent the counterexample discussed above by requiring that the application of the third rule must use the same starting states in the construction of both traces.

Definition 3.13. Let $A = \langle Q, \delta, \gamma, F \rangle$ be a PA and let $U \in \text{Pom}_\Sigma^+$. Suppose that $q_1 \xrightarrow{U}_A q'_1$ and $q_2 \xrightarrow{U}_A q'_2$; these traces are *in lock-step* if one of the following is true:

- (i) $U = a$ for some $a \in \Sigma$
- (ii) $U = V \cdot W$, for V and W smaller than U , and there exist $q''_1, q''_2 \in Q$ such that $q_1 \xrightarrow{V}_A q''_1$ and $q_2 \xrightarrow{V}_A q''_2$ are in lock step, as well as $q''_1 \xrightarrow{W}_A q'_1$ and $q''_2 \xrightarrow{W}_A q'_2$
- (iii) $U = V \parallel W$, for V and W smaller than U , and there exist $r, s \in Q$ and $r', s' \in F$ such that $r \xrightarrow{V}_A r'$ and $s \xrightarrow{W}_A s'$, as well as $q'_1 = \gamma(q_1, \llbracket r, s \rrbracket)$ and $q'_2 = \gamma(q_2, \llbracket r, s \rrbracket)$.

It is easy to see that “being in lock-step” is an equivalence relation on traces.

The lemma below observes that traces that are in lock-step do enjoy determinism.

Lemma 3.14. Let $A = \langle Q, \delta, \gamma, F \rangle$ be a PA and let $q, q'_1, q'_2 \in Q$ and $U \in \text{Pom}_\Sigma^+$. If $q \xrightarrow{U}_A q'_1$ and $q \xrightarrow{U}_A q'_2$ are in lock-step, then $q'_1 = q'_2$.

Proof. The proof proceeds by induction on U . If $U = a \in \Sigma$, then $q'_1 = \delta(q, a)$ and $q'_2 = \delta(q, a)$, and so the claim follows.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , then there exist $q''_1, q''_2 \in Q$ such that $q \xrightarrow{V}_A q''_1$ and $q \xrightarrow{V}_A q''_2$ are in lock-step, as well as $q''_1 \xrightarrow{W}_A q'_1$ and $q''_2 \xrightarrow{W}_A q'_2$. By induction, $q''_1 = q''_2$, and thus again by induction $q'_1 = q'_2$.
- If $U = V \parallel W$ with V and W smaller than U , then there exist $r, s \in Q$ and $r', s' \in F$ such that $r \xrightarrow{V}_A r'$ and $s \xrightarrow{W}_A s'$, and $q'_1 = \gamma(q, \llbracket r, s \rrbracket)$ and $q'_2 = \gamma(q, \llbracket r, s \rrbracket)$. The claim then follows. $\square$

4 Expressions to automata

We now turn our attention to the task of translating a series-rational expression e into a PA that accepts $\llbracket e \rrbracket$. We employ Brzozowski’s method [2] to construct a single *syntactic PA* where every series-rational expression is a state accepting exactly its denotational semantics. To this end we must define which expressions are accepting, and how the sequential and parallel transition functions (*derivatives*, in Brzozowski’s vocabulary) transform states.

We start with the accepting states. In Brzozowski’s original construction, a state is accepting if the denotational semantics of its expression includes the empty word. Analogously, our accepting terms are the terms whose denotational semantics includes the empty pomset.

Definition 4.1. We define the set F_Σ to be the smallest subset of $\mathcal{T}_\Sigma$ satisfying the rules:

$$\frac{}{1 \in F_\Sigma} \quad \frac{e \in F_\Sigma \quad f \in \mathcal{T}_\Sigma}{e + f, f + e \in F_\Sigma} \quad \frac{e, f \in F_\Sigma}{e \cdot f, f \cdot e \in F_\Sigma} \quad \frac{e, f \in F_\Sigma}{e \parallel f, f \parallel e \in F_\Sigma} \quad \frac{e \in \mathcal{T}_\Sigma}{e^* \in F_\Sigma}$$

It is not hard to see that $e \in F_\Sigma$ if and only if $1 \in \llbracket e \rrbracket$. We use $e \star f$ as a shorthand for f if $e \in F_\Sigma$, and $0 \in \mathcal{T}_\Sigma$ otherwise. Let $\mathcal{E}$ be an equation, we write $[\mathcal{E}]$ as a shorthand for $1 \in \mathcal{T}_\Sigma$ if $\mathcal{E}$ holds, and $0 \in \mathcal{T}_\Sigma$ otherwise. We now define sequential and parallel derivatives:

Definition 4.2. We define the functions $\delta_\Sigma : \mathcal{T}_\Sigma \times \Sigma \rightarrow \mathcal{T}_\Sigma$ and $\gamma_\Sigma : \mathcal{T}_\Sigma \times \left(\mathcal{T}_\Sigma^\Sigma\right) \rightarrow \mathcal{T}_\Sigma$ as follows:

$$\begin{array}{ll} \delta_\Sigma(0, a) = 0 & \gamma_\Sigma(0, \phi) = 0 \\ \delta_\Sigma(1, a) = 0 & \gamma_\Sigma(1, \phi) = 0 \\ \delta_\Sigma(b, a) = [a = b] & \gamma_\Sigma(b, \phi) = 0 \\ \delta_\Sigma(e + f, a) = \delta_\Sigma(e, a) + \delta_\Sigma(f, a) & \gamma_\Sigma(e + f, \phi) = \gamma_\Sigma(e, \phi) + \gamma_\Sigma(f, \phi) \\ \delta_\Sigma(e \cdot f, a) = \delta_\Sigma(e, a) \cdot f + e \star \delta_\Sigma(f, a) & \gamma_\Sigma(e \cdot f, \phi) = \gamma_\Sigma(e, \phi) \cdot f + e \star \gamma_\Sigma(f, \phi) \\ \delta_\Sigma(e \parallel f, a) = e \star \delta_\Sigma(f, a) + f \star \delta_\Sigma(e, a) & \gamma_\Sigma(e \parallel f, \phi) = [\phi \simeq \llbracket e, f \rrbracket] + e \star \gamma_\Sigma(f, \phi) + f \star \gamma_\Sigma(e, \phi) \\ \delta_\Sigma(e^*, a) = \delta_\Sigma(e, a) \cdot e^* & \gamma_\Sigma(e^*, \phi) = \gamma_\Sigma(e, \phi) \cdot e^* \end{array}$$

The definition of δ_Σ coincides with Brzozowski's derivative on rational expressions. The definition of γ_Σ mimics the definition of δ_Σ on non-parallel terms except $b \in \Sigma$.

The definition of γ_Σ on parallel terms includes (in the first term) the possibility that the starting states provided to the parallel transition function are (congruent to) the operands of the parallel, in which case the target join state is the accepting state 1. The other two terms (as well as the definition of δ_Σ on a parallel term) account for the fact that if $1 \in \llbracket e \rrbracket$, then $\llbracket f \rrbracket \subseteq \llbracket e \parallel f \rrbracket$. Since we do not allow traces labelled with the empty pomset, traces that originate from these operands are thus lifted to the composition when necessary.

Definition 4.3. The *syntactic PA* is the PA $A_\Sigma = \langle \mathcal{T}_\Sigma, \delta_\Sigma, \gamma_\Sigma, F_\Sigma \rangle$.

We use L_Σ as a shorthand for L_{A_Σ} , and $\rightarrow_\Sigma$ ($\twoheadrightarrow_\Sigma$) as a shorthand for $\rightarrow_{A_\Sigma}$ ($\twoheadrightarrow_{A_\Sigma}$). The remainder of this section is devoted to showing that if $e \in \mathcal{T}_\Sigma$, then $L_\Sigma(e) = \llbracket e \rrbracket$.

4.1 Traces of congruent states

In the analysis of the syntactic trace relation $\rightarrow_\Sigma$, we often encounter sums of terms. To work with these terms, it is often useful to identify sums that are the same modulo associativity, commutativity, idempotence and 0-unit of $+$. In this section, we establish that such an identification is in fact sound, in the sense that if two expressions are related by $\simeq$, then the languages accepted by the states representing those expressions are also identical.

In the first step towards this goal, we show that F_Σ is well-defined with respect to $\simeq$.

Lemma 4.4. Let $e, f \in \mathcal{T}_\Sigma$ be such that $e \simeq f$. Then $e \in F_\Sigma$ if and only if $f \in F_\Sigma$.

Proof. Refer to Appendix A. □

Also, δ_Σ and γ_Σ are well-defined with respect to $\simeq$, in the following sense:

Lemma 4.5. Let $e, f \in \mathcal{T}_\Sigma$ such that $e \simeq f$. If $a \in \Sigma$, then $\delta_\Sigma(e, a) \simeq \delta_\Sigma(f, a)$. Moreover, if $\phi = \{g, h\} \in \binom{\mathcal{T}_\Sigma}{2}$ with $g, h \neq 0$, then $\gamma_\Sigma(e, \phi) \simeq \gamma_\Sigma(f, \phi)$, and if $\psi \in \binom{\mathcal{T}_\Sigma}{2}$ with $\phi \simeq \psi$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e, \psi)$.

Proof. Refer to Appendix A. □

With these lemmas in hand, we can show that $\simeq$ is a “bisimulation” with respect to $\rightarrow_\Sigma$.

Lemma 4.6. Let $e, f \in \mathcal{T}_\Sigma$ be such that $e \simeq f$. If $e \xrightarrow{U}_\Sigma e'$, then there exists an $f' \in \mathcal{T}_\Sigma$ such that $f \xrightarrow{U}_\Sigma f'$ and $e' \simeq f'$.

Proof. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$. But then $e' = \delta_\Sigma(e, a)$. If we choose $f' = \delta_\Sigma(f, a)$, we find that $f \xrightarrow{U}_\Sigma f'$ and $e' \simeq f'$ by Lemma 4.5.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , there exists an $e'' \in \mathcal{T}_\Sigma$ such that $e \xrightarrow{V}_\Sigma e''$ and $e'' \xrightarrow{W}_\Sigma e'$. By induction, we find an $f'' \in \mathcal{T}_\Sigma$ such that $f \xrightarrow{V}_\Sigma f''$ and $e'' \simeq f''$. Again by induction, we find $f' \in \mathcal{T}_\Sigma$ such that $f'' \xrightarrow{W}_\Sigma f'$ and $e' \simeq f'$. This means that $f \xrightarrow{U}_\Sigma f'$ with $e' \simeq f'$.
- If $U = V \parallel W$ with V and W smaller than U , there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{V}_\Sigma g'$ and $h \xrightarrow{W}_\Sigma h'$, and $e' = \gamma_\Sigma(e, \{g, h\})$. Note that, in this case, $g, h \neq 0$ by Lemma 4.7 (which will be proved shortly). If we choose $f' = \gamma_\Sigma(f, \{g, h\})$ we find that $f \xrightarrow{U}_\Sigma f'$ with $e' \simeq f'$ by Lemma 4.5. □

Let I be a finite set, and let $(e_i)_{i \in I}$ be an I -indexed family of terms. In the sequel, we sometimes abuse notation and treat $\sum_{i \in I} e_i$ as a term, where the e_i are summed in some arbitrary order or bracketing. The lemmas above guarantee that the precise choice of representing this sum as a term makes no matter with regard to the traces allowed.

4.2 Trace deconstruction

We proceed with a series of lemmas that characterise reachable states in the syntactic PA. More precisely, we show that expressions of reachable states can be written as sums, each term of which is related to a trace of a subexpression of the starting state. For this reason, we refer to these observations as *trace deconstruction lemmas*: they deconstruct a trace into traces of “smaller” expressions. The purpose of these lemmas is twofold; in Section 4.4, they are used to characterise the languages of expressions as they appear in the syntactic PA, while in Section 4.5 they allow us to bound the reach of an expression.

We start by analysing the traces that originate in base terms, such as 0, 1, or $a \in \Sigma$.

Lemma 4.7. *Let $e, e' \in \mathcal{T}_\Sigma$ and $U \in \text{Pom}_\Sigma^+$ such that $e \xrightarrow{U}_\Sigma e'$. If $e \in \{0, 1\}$, then $e' = 0$. Furthermore, if $e = b \in \Sigma$, then either $e' = 1$ and $U = b$, or $e' = 0$.*

Proof. The proof of the first claim proceeds by induction on U . In the base, $U = a \in \Sigma$, and $e' = \delta_\Sigma(e, a) = 0$, thus the claim holds.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$, with V and W smaller than U , then there exists an $e'' \in \mathcal{T}_\Sigma$ such that $e \xrightarrow{V}_\Sigma e''$ and $e'' \xrightarrow{W}_\Sigma e'$. By induction, $e'' = 0$, and thus again by induction $e' = 0$.
- If $U = V \parallel W$, with V and W smaller than U , then there exist $f, g \in \mathcal{T}_\Sigma$ and $f', g' \in F_\Sigma$ such that $f \xrightarrow{V}_\Sigma f'$ and $g \xrightarrow{W}_\Sigma g'$, and furthermore $e' = \gamma_\Sigma(e, \{f, g\}) = 0$.

The proof of the second claim also proceeds by induction on U . In the base, $U = a \in \Sigma$. Then $e' = \delta_\Sigma(b, a)$. If $a = b$, then $e' = 1$, otherwise $e' = 0$; thus the claim holds.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , then there exists an $e'' \in \mathcal{T}_\Sigma$ such that $b \xrightarrow{V}_\Sigma e''$ and $e'' \xrightarrow{W}_\Sigma e'$. By induction, either $V = b$ and $e'' = 1$, or $e'' = 0$. In either case, $e' = 0$ by the first claim.
- If $U = V \parallel W$ with V and W smaller than U , then there exist $f, g \in \mathcal{T}_\Sigma$ and $f', g' \in F_\Sigma$ such that $f \xrightarrow{V}_\Sigma f'$ and $g' \xrightarrow{W}_\Sigma g'$, and $e' = \gamma_\Sigma(e, \{f, g\}) = 0$. The claim thus holds immediately. $\square$

Note, however, that the first claim above does not imply that 0 and 1 are indistinguishable as far as traces are concerned, for $0 \notin F_\Sigma$ while $1 \in F_\Sigma$.

We also consider the traces that originate in a sum of terms. The intuition here is that the input is processed by both terms simultaneously, and thus the target state must be the sum of the states that are the result of processing the input for to each term individually.

Lemma 4.8. *Let $e_1, e_2 \in \mathcal{T}_\Sigma$ and $U \in \text{Pom}_\Sigma^+$. If $e_1 + e_2 \xrightarrow{U}_\Sigma e'$, then there exist $e'_1, e'_2 \in \mathcal{T}_\Sigma$ such that $e' = e'_1 + e'_2$, and $e_1 \xrightarrow{U}_\Sigma e'_1$ and $e_2 \xrightarrow{U}_\Sigma e'_2$.*

Proof. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$, and $e' = \delta_\Sigma(e_1 + e_2, a) = \delta_\Sigma(e_1, a) + \delta_\Sigma(e_2, a)$. We can then choose $e'_1 = \delta_\Sigma(e_1, a)$ and $e'_2 = \delta_\Sigma(e_2, a)$ to validate the claim.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , then there exists an $e'' \in \mathcal{T}_\Sigma$ such that $e_1 + e_2 \xrightarrow{V}_\Sigma e''$ and $e'' \xrightarrow{W}_\Sigma e'$. By induction, we find $e''_1, e''_2 \in \mathcal{T}_\Sigma$ such that $e'' = e''_1 + e''_2$, and $e_1 \xrightarrow{V}_\Sigma e''_1$ and $e_2 \xrightarrow{V}_\Sigma e''_2$. Again by induction, we find $e'_1, e'_2 \in \mathcal{T}_\Sigma$ such that $e' = e'_1 + e'_2$, and $e''_1 \xrightarrow{W}_\Sigma e'_1$ and $e''_2 \xrightarrow{W}_\Sigma e'_2$. In conclusion, we find that $e_1 \xrightarrow{U}_\Sigma e'_1$ and $e_2 \xrightarrow{U}_\Sigma e'_2$.
- If $U = V \parallel W$, with V and W smaller than U , then there exist $f, g \in \mathcal{T}_\Sigma$ such that $e' = \gamma_\Sigma(e_1 + e_2, \{f, g\}) = \gamma_\Sigma(e_1, \{f, g\}) + \gamma_\Sigma(e_2, \{f, g\})$, and furthermore there exist $f', g' \in F_\Sigma$ such that $f \xrightarrow{V}_\Sigma f'$ and $g \xrightarrow{W}_\Sigma g'$. We can choose $e'_1 = \gamma_\Sigma(e_1, \{f, g\})$ and $e'_2 = \gamma_\Sigma(e_2, \{f, g\})$ to validate the claim. $\square$

We now consider the traces starting in a state that is a sequential composition. The intuition here is that the syntactic PA must first proceed through the left operand, before it can proceed to process the right operand. Thus, either the pomset is processed by the left operand entirely, or we should be able to split the pomset in two sequential parts: the first part is processed by the left operand, and the second by the right operand.

Lemma 4.9. *Let $e_1, e_2 \in \mathcal{T}_\Sigma$ and $U \in \text{Pom}_\Sigma^+$ be such that $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$. There exist an $f' \in \mathcal{T}_\Sigma$ and a finite set I , as well as I -indexed families of terms $(f'_i)_{i \in I} \in F_\Sigma$ and $(f_i)_{i \in I} \in \mathcal{T}_\Sigma$ and of pomsets $(U'_i)_{i \in I} \in \text{Pom}_\Sigma$ and $(U_i)_{i \in I} \in \text{Pom}_\Sigma^+$, such that:*

- $f \simeq f' \cdot e_2 + \sum_{i \in I} f_i$ and $e_1 \xrightarrow{U}_\Sigma f'$
- for all $i \in I$, $e_1 \xrightarrow{U'_i}_\Sigma f'_i$, $e_2 \xrightarrow{U_i}_\Sigma f_i$, and $U = U'_i \cdot U_i$.

Proof. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$, and $f = \delta_\Sigma(e_1 \cdot e_2, a)$. If on the one hand $e_1 \notin F_\Sigma$, then $f \simeq \delta_\Sigma(e_1, a) \cdot e_2$; we choose $f' = \delta_\Sigma(e_1, a)$ and $I = \emptyset$. If on the other hand $e_1 \in F_\Sigma$, then $f \simeq \delta_\Sigma(e_1, a) \cdot e_2 + \delta_\Sigma(e_2, a)$; we choose $f' = \delta_\Sigma(e_1, a)$, $I = \{*\}$, $f'_* = e_1$, $f_* = \delta_\Sigma(e_2, a)$, $U'_* = 1$ and $U_* = U$. In either case, our choices validate the claim.

For the inductive step, we consider two cases.

- If $U = V \cdot W$, with V and W smaller than U , then there exists a g such that $e_1 \cdot e_2 \xrightarrow{V}_\Sigma g$ and $g \xrightarrow{W}_\Sigma f$. By induction, we obtain a $g' \in \mathcal{T}_\Sigma$ and a finite set J , as well as J -indexed families of terms $(g'_j)_{j \in J} \in F_\Sigma$ and $(g_j)_{j \in J} \in \mathcal{T}_\Sigma$ and of pomsets $(V'_j)_{j \in J} \in \text{Pom}_\Sigma$ and $(V_j)_{j \in J} \in \text{Pom}_\Sigma^+$, such that

- $g \simeq g' \cdot e_2 + \sum_{j \in J} g_j$ and $e_1 \xrightarrow{V}_\Sigma g'$
- for all $j \in J$, $e_1 \xrightarrow{V'_j}_\Sigma g'_j$, $e_2 \xrightarrow{V_j}_\Sigma g_j$ and $V = V'_j \cdot V_j$.

By Lemma 4.6 and $g \simeq g' \cdot e_2 + \sum_{j \in J} g_j$, as well as Lemma 4.8 and $g \xrightarrow{W}_\Sigma f$, there exist $h \in \mathcal{T}_\Sigma$ and a J -indexed family of terms $(d_j)_{j \in J} \in F_\Sigma$ such that $f \simeq h + \sum_{j \in J} d_j$, $g' \cdot e_2 \xrightarrow{W}_\Sigma h$ and for all $j \in J$, also $g_j \xrightarrow{W}_\Sigma d_j$.

Since $g' \cdot e_2 \xrightarrow{W}_\Sigma h$, we can find (again by induction) an $h' \in \mathcal{T}_\Sigma$, and a finite set K (without loss of generality, disjoint from J) and K -indexed families of terms $(h'_k)_{k \in K} \in F_\Sigma$ and $(h_k)_{k \in K} \in \mathcal{T}_\Sigma$ and of pomsets $(W'_k)_{k \in K} \in \text{Pom}_\Sigma$ and $(W_k)_{k \in K} \in \text{Pom}_\Sigma^+$ such that

- $h \simeq h' \cdot e_2 + \sum_{k \in K} h_k$ and $g' \xrightarrow{W}_\Sigma h'$
- for all $k \in K$, $g' \xrightarrow{W'_k}_\Sigma h'_k$, $e_2 \xrightarrow{W_k}_\Sigma h_k$ and $W = W'_k \cdot W_k$.

We now choose $I = J \cup K$, and $f' = h'$. Furthermore, we choose I -indexed families of terms $(f'_i)_{i \in I} \in F_\Sigma$ and $(f_i)_{i \in I} \in \mathcal{T}_\Sigma$ and of pomsets $(U'_i)_{i \in I} \in \text{Pom}_\Sigma$ and $(U_i)_{i \in I} \in \text{Pom}_\Sigma^+$ as follows:

$$f'_i = \begin{cases} g'_i & i \in J \\ h'_i & i \in K \end{cases} \quad f_i = \begin{cases} d_i & i \in J \\ h_i & i \in K \end{cases} \quad U'_i = \begin{cases} V'_i & i \in J \\ V \cdot W'_i & i \in K \end{cases} \quad U_i = \begin{cases} V_i \cdot W & i \in J \\ W_i & i \in K \end{cases}$$

It remains to verify the requirements on our choices one by one. For the first claim, we can derive

$$f \simeq h + \sum_{j \in J} d_j \simeq h' \cdot e_2 + \sum_{k \in K} h_k + \sum_{j \in J} d_j \simeq f' \cdot e_2 + \sum_{i \in I} f_i$$

Furthermore, since $e_1 \xrightarrow{V}_\Sigma g'$ and $g' \xrightarrow{W}_\Sigma h'$, we have that $e_1 \xrightarrow{U}_\Sigma h' = f'$. For the second claim, let us fix an $i \in I$. Suppose first that $i \in J$, then:

- $e_1 \xrightarrow{V'_i}_\Sigma g'_i$, thus $e_1 \xrightarrow{U'_i}_\Sigma f'_i$
- $e_2 \xrightarrow{V_i}_\Sigma g_i$ and $g_i \xrightarrow{W}_\Sigma d_i$, thus $e_2 \xrightarrow{U_i}_\Sigma f_i$
- $U'_i \cdot U_i = V'_i \cdot V_i \cdot W = V \cdot W = U$.

Secondly, suppose that $i \in K$, then:

- $e_1 \xrightarrow{V}_\Sigma g'$ and $g' \xrightarrow{W'_i}_\Sigma h'_i$, thus $e_1 \xrightarrow{U'_i}_\Sigma f'_i$
- $e_2 \xrightarrow{W_i}_\Sigma h_i$, thus $e_2 \xrightarrow{U_i}_\Sigma f_i$
- $U'_i \cdot U_i = V \cdot W'_i \cdot W_i = V \cdot W = U$.

All requirements are thus validated for our choices in this case.

- If $U = V \parallel W$, then there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{V}_\Sigma g'$ and $h \xrightarrow{W}_\Sigma h'$, and we know that $f = \gamma_\Sigma(e_1 \cdot e_2, \{g, h\})$. If on the one hand $e_1 \notin F_\Sigma$, then $f \simeq \gamma_\Sigma(e_1, \{g, h\}) \cdot e_2$; we choose $f' = \gamma_\Sigma(e_1, \{g_1, g_2\})$ and $I = \emptyset$. If on the other hand $e_1 \in F_\Sigma$, then $f \simeq \gamma_\Sigma(e_1, \{g, h\}) \cdot e_2 + \gamma_\Sigma(e_2, \{g, h\})$; we choose $f' = \gamma_\Sigma(e_1, \{g_1, g_2\})$ and $I = 1$, as well as $f'_* = e_1$, $f_* = \gamma_\Sigma(e_2, \{g_1, g_2\})$, $U'_* = 1$ and $U'_* = U$. In both cases, our choices validate the claim. $\square$

The next deconstruction lemma concerns traces originating in a parallel composition. Intuitively, the syntactic PA either processes parallel components of the pomset, or processes according to one operand, provided that the other operand allows immediate acceptance.

Lemma 4.10. *If $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$, then there exist $f_1, f_2, f_3 \in \mathcal{T}_\Sigma$, such that*

- $f \simeq f_1 + f_2 + f_3$
- either $f_1 = 0$, or $e_2 \in F_\Sigma$ and $e_1 \xrightarrow{U}_\Sigma f_1$
- either $f_2 = 0$, or $e_1 \in F_\Sigma$ and $e_2 \xrightarrow{U}_\Sigma f_2$
- either $f_3 = 0$, or $f_3 = 1$ and there exist $e'_1, e'_2 \in \mathcal{T}_\Sigma$ and $f'_1, f'_2 \in F_\Sigma$ and $U_1, U_2 \in \mathbf{Pom}_\Sigma^+$ such that $e_1 \simeq e'_1$ and $e_2 \simeq e'_2$ and $U = U_1 \parallel U_2$ and $e'_1 \xrightarrow{U_1}_\Sigma f'_1$ and $e'_2 \xrightarrow{U_2}_\Sigma f'_2$.

Proof. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$. Now, choose $f_2 = e_1 \star \delta_\Sigma(e_2, a)$ and $f_1 = e_2 \star \delta_\Sigma(e_1, a)$. We also choose $f_3 = 0$. It is easy to validate that the claim holds for these choices.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , then there exists a $g \in \mathcal{T}_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{V}_\Sigma g$ and $g \xrightarrow{W}_\Sigma f$. By induction, we obtain g_1, g_2, g_3 such that:
 - $g \simeq g_1 + g_2 + g_3$
 - either $g_1 = 0$, or $e_2 \in F_\Sigma$ and $e_1 \xrightarrow{V}_\Sigma g_1$
 - either $g_2 = 0$, or $e_1 \in F_\Sigma$ and $e_2 \xrightarrow{V}_\Sigma g_2$
 - either $g_3 = 0$, or $g_3 = 1$ and there exist $e'_1, e'_2 \in \mathcal{T}_\Sigma$ and $g'_1, g'_2 \in F_\Sigma$ and $V_1, V_2 \in \mathbf{Pom}_\Sigma^+$ such that $e_1 \simeq e'_1$ and $e_2 \simeq e'_2$ and $V = V_1 \parallel V_2$ and $e'_1 \xrightarrow{V_1}_\Sigma g'_1$ and $e'_2 \xrightarrow{V_2}_\Sigma g'_2$.

By $g \simeq g_1 + g_2 + g_3$ and Lemma 4.6, as well as $g \xrightarrow{W}_\Sigma f$ and Lemma 4.8, we obtain $f_1, f_2, f_3 \in \mathcal{T}_\Sigma$ such that $f \simeq f_1 + f_2 + f_3$, and $g_1 \xrightarrow{W}_\Sigma f_1$, $g_2 \xrightarrow{W}_\Sigma f_2$ and $g_3 \xrightarrow{W}_\Sigma f_3$. We now validate the remaining claims. First, note that if $g_i = 0$ for all $i \in \{1, 2, 3\}$, then $f_i = 0$ by Lemma 4.7. As to the remaining possibilities:

- If $g_1 \neq 0$, then $e_2 \in F_\Sigma$ and by $e_1 \xrightarrow{V}_\Sigma g_1$ and $g_1 \xrightarrow{W}_\Sigma f_1$ we conclude that $e_1 \xrightarrow{U}_\Sigma f_1$.
- If $g_2 \neq 0$, then $e_1 \in F_\Sigma$ and by $e_2 \xrightarrow{V}_\Sigma g_2$ and $g_2 \xrightarrow{W}_\Sigma f_2$ we conclude that $e_2 \xrightarrow{U}_\Sigma f_2$.
- If $g_3 \neq 0$, then $g_3 = 1$ and we conclude that $f_3 = 0$ by Lemma 4.7.
- If $U = V \parallel W$ with V and W smaller than U , then there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $f = \gamma_\Sigma(e_1 \parallel e_2, \{g, h\})$, and $g \xrightarrow{V}_\Sigma g' \in F_\Sigma$ and $h \xrightarrow{W}_\Sigma h' \in F_\Sigma$. We choose $f_1 = e_2 \star \gamma_\Sigma(e_1, \{g, h\})$ and $f_2 = e_1 \star \gamma_\Sigma(e_2, \{g, h\})$. Furthermore, we set $f_3 = [\{e_1, e_2\} = \{g, h\}]$. It is now easy to see that $f \simeq f_1 + f_2 + f_3$. We validate the remaining claims.
 - If $e_2 \notin F_\Sigma$, then $f_1 = 0$. Otherwise, $e_2 \in F_\Sigma$ and $e_1 \xrightarrow{U}_\Sigma \gamma_\Sigma(e_1, \{g, h\}) = f_1$.
 - If $e_1 \notin F_\Sigma$, then $f_2 = 0$. Otherwise, $e_1 \in F_\Sigma$ and $e_2 \xrightarrow{U}_\Sigma \gamma_\Sigma(e_2, \{g, h\}) = f_2$.
 - If $\{e_1, e_2\} \neq \{g, h\}$, then $f_3 = 0$. Otherwise, assume (without loss of generality) that $e_1 \simeq g$ and $e_2 \simeq h$; we can choose $e'_1 = g$ and $e'_2 = h$, $f'_1 = g' \in F_\Sigma$ and $f'_2 = h' \in F_\Sigma$, as well as $U_1 = V \in \mathbf{Pom}_\Sigma^+$ and $U_2 = W \in \mathbf{Pom}_\Sigma^+$ to find that $g \xrightarrow{U_1}_\Sigma f'_1$ and $h \xrightarrow{U_2}_\Sigma f'_2$. $\square$

Finally, we analyse the reachable states of an expression of the form e^* . The intuition here is that, starting in e^* , the PA can process traces originating in e indefinitely. The trace should thus be sequentially decomposable, with each component the label of a trace originating in e . Furthermore, all but the last target state of these traces should be accepting.

Lemma 4.11. *If $e^* \xrightarrow{U}_\Sigma f$, then there exists a finite set I and an I -indexed family of finite sets $(J_i)_{i \in I}$, as well as I -indexed families of terms $(f_i)_{i \in I} \in \mathcal{T}_\Sigma$ and of pomsets $(U_i)_{i \in I} \in \text{Pom}_\Sigma^+$, and for all $i \in I$ also J_i -indexed families of terms $(f_{i,j})_{j \in J_i} \in F_\Sigma$ and of pomsets $(U_{i,j})_{j \in J_i} \in \text{Pom}_\Sigma^+$, such that $f \simeq \sum_{i \in I} f_i \cdot e^*$, and for all $i \in I$:*

- $e \xrightarrow{U_i}_\Sigma f_i$
- for all $j \in J_i$ we have that $e \xrightarrow{U_{i,j}}_\Sigma f_{i,j}$
- $U = U'_i \cdot U_i$, where U'_i is some concatenation of all $U_{i,j}$ for all $j \in J_i$.

Proof. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$, and so $f = \delta_\Sigma(e^*, a) = \delta_\Sigma(e, a) \cdot e^*$. We can choose $I = \{*\}$, $J_* = \emptyset$ and $U_* = a$ to validate the claim.

For the inductive step, we consider two cases.

- If $U = V \cdot W$, with V and W smaller than U , there exists a g such that $e^* \xrightarrow{V}_\Sigma g$ and $g \xrightarrow{W}_\Sigma f$. Since the remainder of this part of the proof is somewhat involved, we begin by outlining our strategy. First, we deconstruct the trace $e^* \xrightarrow{V}_\Sigma g$, by induction. Then we deconstruct the trace $g \xrightarrow{W}_\Sigma f$, making use of the fact that g can be seen a sum of the form found in the claim, i.e., where each term is of the form $g' \cdot e^*$, and thus (by Lemma 4.8) f can be seen as a sum of terms f' such that $g' \cdot e^* \xrightarrow{W}_\Sigma f'$. We then leverage Lemma 4.9 to deconstruct each of the latter traces. In the end, we have a big cache of variables, which we use to construct the families of terms and pomsets required by the claim. Since we will be dealing with a fair number of index sets, we tacitly assume (without loss of generality) that of them are disjoint.

As for the proof, consider the trace $e^* \xrightarrow{V}_\Sigma g$. By induction, we obtain a finite set I' and an I' -indexed family of finite sets $(J'_i)_{i \in I'}$, as well as I' -indexed families of terms $(g_i)_{i \in I'} \in \mathcal{T}_\Sigma$ and pomsets $(V_i)_{i \in I'} \in \text{Pom}_\Sigma^+$ and for all $i \in I'$ also J'_i -indexed families of terms $(g_{i,j})_{j \in J'_i} \in F_\Sigma$ and pomsets $(V_{i,j})_{j \in J'_i} \in \text{Pom}_\Sigma^+$, such that $g \simeq \sum_{i \in I'} g_i \cdot e^*$, and for all $i \in I'$:

- $e \xrightarrow{V_i}_\Sigma g_i$
- for all $j \in J'_i$ we have that $e \xrightarrow{V_{i,j}}_\Sigma g_{i,j} \in F_\Sigma$
- $V = V'_i \cdot V_i$, where V'_i is some concatenation of $V_{i,j}$ for all $j \in J'_i$.

By the fact that $g \simeq \sum_{i \in I'} g_i \cdot e^*$ and by Lemma 4.6, as well as $g \xrightarrow{W}_\Sigma f$ and Lemma 4.8, we find an I' -indexed family of terms $(h_i)_{i \in I'} \in \mathcal{T}_\Sigma$ such that $f \simeq \sum_{i \in I'} h_i$, and for all $i \in I'$ also $g_i \cdot e^* \xrightarrow{W}_\Sigma h_i$. Then, by Lemma 4.9, we find for all $i \in I'$ a term h'_i and finite set K_i as well as K_i -indexed families of terms $(h'_{i,k})_{k \in K_i} \in F_\Sigma$ and $(h_{i,k})_{k \in K_i} \in \mathcal{T}_\Sigma$ and of pomsets $(W'_{i,k})_{k \in K_i} \in \text{Pom}_\Sigma$ and $(W_{i,k})_{k \in K_i} \in \text{Pom}_\Sigma^+$, such that

- $h_i \simeq h'_i \cdot e^* + \sum_{k \in K_i} h_{i,k}$ and $g_i \xrightarrow{W}_{\Sigma} h'_i$
- for all $k \in K_i$, $g_i \xrightarrow{W'_{i,k}}_\Sigma h'_{i,k}$, $e^* \xrightarrow{W_{i,k}}_\Sigma h_{i,k}$ and $W = W'_k \cdot W_k$.

By induction and since for all $i \in I'$ and $k \in K_i$ we have that $e^* \xrightarrow{W_{i,k}}_\Sigma h_{i,k}$, we obtain for this i and k a finite set $L_{i,k}$ and an $L_{i,k}$ -indexed family of finite sets $(M_{i,k,\ell})_{\ell \in L_{i,k}}$, as well as $L_{i,k}$ -indexed families of terms $(h_{i,k,\ell})_{\ell \in L_{i,k}} \in \mathcal{T}_\Sigma$ and of pomsets $(W_{i,k,\ell})_{\ell \in L_{i,k}} \in \text{Pom}_\Sigma^+$, and for all $\ell \in L_{i,k}$ also $M_{i,k,\ell}$ -indexed families of terms $(h_{i,k,\ell,m})_{m \in M_{i,k,\ell}} \in F_\Sigma$ and of pomsets $(V_{i,k,\ell,m})_{m \in M_{i,k,\ell}} \in \text{Pom}_\Sigma^+$ such that $h_{i,k} \simeq \sum_{\ell \in L_{i,k}} h_{i,k,\ell} \cdot e^*$, and for all $\ell \in L_{i,k}$:

- $e \xrightarrow{W_{i,k,\ell}}_\Sigma h_{i,k,\ell}$
- for all $m \in M_{i,k,\ell}$ we have that $e \xrightarrow{W_{i,k,\ell,m}}_\Sigma h_{i,k,\ell,m}$
- $W_i = W'_{i,k,\ell} \cdot W_{i,k,\ell}$, where $W'_{i,k,\ell}$ is some concatenation of $W_{i,k,\ell,m}$ for $m \in M_{i,k,\ell}$.

We are now ready to choose the required (families of) sets, terms and pomsets, as follows:

- $I = I' + \bigcup_{k \in K_i} L_{i,k}$

– for all $i \in I$, we set

$$J_i = \begin{cases} J'_i & i \in I' \\ M_{i',k,i} \cup \{*_i\} & i \in L_{i',k}, k \in K_{i'} \end{cases} \quad f_i = \begin{cases} h'_i & i \in I' \\ h_{i',k,i} & i \in L_{i',k}, k \in K_{i'} \end{cases}$$

where $*_i$ is a “fresh” symbol not in any index set. We furthermore choose

$$U_i = \begin{cases} V_i \cdot W & i \in I' \\ W_{i',k,i} & i \in L_{i',k}, k \in K_{i'} \end{cases}$$

– for all $i \in I$ and $j \in J_i$, we set

$$f_{i,j} = \begin{cases} g_{i,j} & i \in I', j \in J'_i \\ h'_{i',k} & i \in L_{i',k}, k \in K_{i'}, j = *_i \\ h_{i',k,i,j} & i \in L_{i',k}, k \in K_{i'}, j \neq *_i \end{cases}$$

and furthermore

$$U_{i,j} = \begin{cases} V_{i,j} & i \in I', j \in J'_i \\ V_{i'} \cdot W'_{i'} & i \in L_{i',k}, k \in K_{i'}, j = *_i \\ W_{i',k,i,j} & i \in L_{i',k}, k \in K_{i'}, j \neq *_i \end{cases}$$

It remains to check the requirements on our choices.

– One easily verifies that

$$f \simeq \sum_{i \in I'} h_i \simeq \sum_{i \in I'} \left(h'_i \cdot e^* + \sum_{i \in K_i} h_{i,k} \right) \simeq \sum_{i \in I'} \left(h'_i \cdot e^* + \sum_{k \in K_i} \sum_{\ell \in L_{i,k}} h_{i,k,\ell} \cdot e^* \right) \simeq \sum_{i \in I} f_i \cdot e^*$$

– If $i \in I$, there are two cases to consider.

* If $i \in I'$, then since $e \xrightarrow{V_i}_{\rightarrow \Sigma} g_i$ and $g_i \xrightarrow{W}_{\rightarrow \Sigma} h'_i$, we find that $e \xrightarrow{U_i}_{\rightarrow \Sigma} f_i$.

* If $i \in L_{i',k}$ for some $i' \in I'$ and $k \in K_{i'}$, then since $e \xrightarrow{W_{i',k,i}}_{\rightarrow \Sigma} h_{i',k,i}$ we find that $e \xrightarrow{U_i}_{\rightarrow \Sigma} f_i$.

– If $i \in I$ and $j \in J_i$, there are three cases to consider.

* If $i \in I'$ and $j \in J'_i$, then since $e \xrightarrow{V_{i,j}}_{\rightarrow \Sigma} g_{i,j}$, we find that $e \xrightarrow{U_{i,j}}_{\rightarrow \Sigma} f_{i,j}$.

* If $i \in L_{i',k}$ for some $i' \in I'$ and $k \in K_{i'}$, and $j = *_i$, then since $e \xrightarrow{V_{i'}}_{\rightarrow \Sigma} g_i$ and $g_i \xrightarrow{W'_{i'}}_{\rightarrow \Sigma} h'_{i',k}$, we find that $e \xrightarrow{U_{i,j}}_{\rightarrow \Sigma} f_{i,j}$.

* If $i \in L_{i',k}$ for some $i' \in I'$ and $k \in K_{i'}$, and $j \neq *_i$ (thus $j \in M_{i',k,i}$), then since $e \xrightarrow{W_{i',k,i,j}}_{\rightarrow \Sigma} h_{i',k,i,j}$ we find that $e \xrightarrow{U_{i,j}}_{\rightarrow \Sigma} f_{i,j}$.

– If $i \in I$, there are two cases to consider.

* If $i \in I'$, then $V = V'_i \cdot V_i$, where V'_i is some concatenation of $V_{i,j}$ for all $j \in J'_i$. But then we can choose $U'_i = V'_i$ as a concatenation of $U_{i,j}$ for all $j \in J_i$, to find that $U'_i \cdot U_i = V'_i \cdot V_i \cdot W = V \cdot W = U$.

* If $i \in L_{i',k}$ for all $i' \in I'$ and $k \in K_{i'}$, then we know that there exists a pomset $W'_{i',k,i}$ which is a concatenation of $W_{i',k,i,m}$ for all $m \in M_{i',k,i}$, such that $W'_{i',k,i} \cdot W_{i',k,i} = W'_{i'}$. We can then choose $U'_i = V \cdot W'_{i'} \cdot W'_{i',k,i}$ as a concatenation of $U_{i,j}$ with $j \in J_i$, and find that $U'_i \cdot U_i = V \cdot W'_{i'} \cdot W'_{i',k,i} \cdot W_{i',k,i} = V \cdot W'_{i'} \cdot W_{i'} = V \cdot W = U$.

- If $U = V \parallel W$, then there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in \mathcal{T}_\Sigma$ such that $g \xrightarrow{V}_{\rightarrow \Sigma} g'$ and $h \xrightarrow{W}_{\rightarrow \Sigma} h'$, and $f = \gamma_\Sigma(e^*, \llbracket g, h \rrbracket) = \gamma_\Sigma(e, \llbracket g, h \rrbracket) \cdot e^*$. In this case, we choose $I = \{*\}$, and $J_* = \emptyset$ and $U_* = U$ to find that the claims are validated. $\square$

4.3 Trace construction

In the above, we learned how to deconstruct traces in the syntactic PA. To verify that the syntactic PA indeed accepts any series-rational pomset language, we also need to show the reverse, that is, how to *construct* traces in the syntactic PA from smaller traces. In this context it is often useful to work with the preorder obtained from $\simeq$.

Definition 4.12. The relation $\lesssim \subseteq \mathcal{T}_\Sigma \times \mathcal{T}_\Sigma$ is defined by $e \lesssim f$ if and only if $e + f \simeq f$.

The intuition to $e \lesssim f$ is that e consists of one or more terms that also appear in f , up to $\simeq$.

In analogy to Lemma 4.6, we show that $\lesssim$ is a “simulation” with respect to traces.

Lemma 4.13. Let $e, e', f \in \mathcal{T}_\Sigma$ be such that $e \lesssim f$. If $e \xrightarrow{U}_\Sigma e'$, then there exists an $f' \in \mathcal{T}_\Sigma$ such that $f \xrightarrow{U}_\Sigma f'$ and $e' \lesssim f'$. Furthermore, if $e \in F_\Sigma$, then $f \in F_\Sigma$.

Proof. We prove the first claim by induction on U . In the base, $U = a \in \Sigma$ and $e' = \delta_\Sigma(e, a)$. Note that $\delta_\Sigma(e, a) + \delta_\Sigma(f, a) = \delta_\Sigma(e + f, a) \simeq \delta_\Sigma(f, a)$ by Lemma 4.5. We choose $f' = \delta_\Sigma(f, a)$ to find that $f \xrightarrow{U}_\Sigma f'$ with $e' = \delta_\Sigma(e, a) \lesssim \delta_\Sigma(f, a) = f'$.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , then there exists an e'' such that $e \xrightarrow{V}_\Sigma e''$ and $e'' \xrightarrow{W}_\Sigma e'$. By induction, we find an $f'' \in \mathcal{T}_\Sigma$ such that $f \xrightarrow{V}_\Sigma f''$ and $e'' \lesssim f''$, and again by induction we find an $f' \in \mathcal{T}_\Sigma$ such that $f'' \xrightarrow{W}_\Sigma f'$ with $e' \lesssim f'$. In total, we know that $e \xrightarrow{U}_\Sigma f$ with $e' \lesssim f'$.
- If $U = V \parallel W$ with V and W smaller than U , then there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{V}_\Sigma g'$ and $h \xrightarrow{W}_\Sigma h'$ and $e' = \gamma_\Sigma(e, \llbracket g, h \rrbracket)$. Note that $\gamma_\Sigma(e, \llbracket g, h \rrbracket) + \gamma_\Sigma(f, \llbracket g, h \rrbracket) = \gamma_\Sigma(e + f, \llbracket g, h \rrbracket) \simeq \gamma_\Sigma(f, \llbracket g, h \rrbracket)$ by Lemma 4.5. We choose $f' = \gamma_\Sigma(f, \llbracket g, h \rrbracket)$ to find that $f \xrightarrow{U}_\Sigma f'$ with $e' = \gamma_\Sigma(e, \phi) \lesssim \gamma_\Sigma(f, \phi) = f'$.

For the second claim, suppose that $e \in F_\Sigma$, then also $e + f \in F_\Sigma$. But then $f \in F_\Sigma$ by Lemma 4.4. $\square$

The following lemma tells us that we can create a trace labelled with the concatenation of the labels of two smaller traces, and starting in the sequential composition of the original starting states, provided that the first trace ends in an accepting state. Furthermore, the target state of the newly constructed trace contains the target state of the second trace. We also prove two auxiliary claims towards this end, which will be useful later on.

Lemma 4.14. Let $e_1, e_2, f_1, f_2 \in \mathcal{T}_\Sigma$ and $U, V \in \text{Pom}_\Sigma^+$ be such that $e_1 \xrightarrow{U}_\Sigma f_1$ and $e_2 \xrightarrow{V}_\Sigma f_2$. The following hold:

- There exists an $f \in \mathcal{T}_\Sigma$ such that $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e_2 \lesssim f$.
- If $f_1 \in F_\Sigma$, then there exists an $f \in \mathcal{T}_\Sigma$ such that $f_1 \cdot e_2 \xrightarrow{V}_\Sigma f$ with $f_2 \lesssim f$.
- If $f_1 \in F_\Sigma$, then there exists an $f \in \mathcal{T}_\Sigma$ such that $e_1 \cdot e_2 \xrightarrow{U \cdot V}_\Sigma f$ with $f_2 \lesssim f$.

Proof. The proof of the first claim proceeds by induction on U . In the base, $U = a \in \Sigma$ and $f_1 = \delta_\Sigma(e_1, a)$. We choose $f = \delta_\Sigma(e_1 \cdot e_2, a)$. It is now easy to show that $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e_2 \lesssim f$.

For the inductive step, there are two cases to consider.

- If $U = W \cdot X$ with W and X smaller than U , there exists a g_1 such that $e_1 \xrightarrow{W}_\Sigma g_1$ and $g_1 \xrightarrow{X}_\Sigma f_1$. By induction, we obtain $g \in \mathcal{T}_\Sigma$ such that $e_1 \cdot e_2 \xrightarrow{W}_\Sigma g$ with $g_1 \cdot e_2 \lesssim g$. Again by induction, we find $h \in \mathcal{T}_\Sigma$ such that $g_1 \cdot e_2 \xrightarrow{X}_\Sigma h$ with $f_1 \cdot e_2 \lesssim h$. By Lemma 4.13, we then know that $g \xrightarrow{X}_\Sigma f$ for some f with $h \lesssim f$. In total, we find that $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e_2 \lesssim f$ (by transitivity of $\lesssim$).
- If $U = W \parallel X$ with W and X smaller than U , there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{W}_\Sigma g'$ and $h \xrightarrow{X}_\Sigma h'$, and $f_1 = \gamma_\Sigma(e_1, \llbracket g, h \rrbracket)$. We choose $f = \gamma_\Sigma(e_1 \cdot e_2, \llbracket g, h \rrbracket)$. It is now easy to show that $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e_2 \lesssim f$.

The proof of the second claim proceeds by induction on V . In the base, $V = a \in \Sigma$ and $f_2 = \delta_\Sigma(e_2, a)$. We can then choose $f = \delta_\Sigma(f_1 \cdot e_2, a)$. It is now easy to show that $f_1 \cdot e_2 \xrightarrow{V}_\Sigma f$ with $f_2 \lesssim f$.

For the inductive step, there are two cases to consider.

- If $V = W \cdot X$ with W and X smaller than V , there exists a $g_2 \in \mathcal{T}_\Sigma$ such that $e_2 \xrightarrow{W}_\Sigma g_2$ and $g_2 \xrightarrow{X}_\Sigma f_2$. By induction, there exists a $g \in \mathcal{T}_\Sigma$ such that $f_1 \cdot e_2 \xrightarrow{W}_\Sigma g$ with $g_2 \lesssim g$. By Lemma 4.13, we find f such that $g \xrightarrow{X}_\Sigma f$ with $f_2 \lesssim f$, and thus $f_1 \cdot e_2 \xrightarrow{V}_\Sigma f$.
- If $V = W \parallel X$ with W and X smaller than V , there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{W}_\Sigma g'$ and $h \xrightarrow{X}_\Sigma h'$ and $f_2 = \gamma_\Sigma(e_2, \llbracket g, h \rrbracket)$. We can then choose $f = \gamma_\Sigma(f_1 \cdot e_2, \llbracket g, h \rrbracket)$. It is now easy to show that $f_1 \cdot e_2 \xrightarrow{V}_\Sigma f$ with $f_2 \lesssim f$.

The third claim is a direct consequence of the first two claims and Lemma 4.13. $\square$

We can also construct traces that start in a parallel composition. One way is to construct traces that start in each operand and reach an accepting state; we obtain a trace in their parallel composition almost trivially. If one of the operands is accepting, we can also construct a single trace that starts in the other operand and obtain a trace with the same label starting in the parallel construction. In both cases, we describe the target of the new trace using $\lesssim$.

Lemma 4.15. *Let $e_1, e_2 \in \mathcal{T}_\Sigma$. The following hold:*

- If $f_1, f_2 \in F_\Sigma$ and $U, V \in \mathbf{Pom}_\Sigma^+$ are such that $e_1 \xrightarrow{U}_\Sigma f_1$ and $e_2 \xrightarrow{V}_\Sigma f_2$, then there exists an $f \in \mathcal{T}_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{U \parallel V}_\Sigma f$ with $1 \lesssim f$.
- If $e_2 \in F_\Sigma$ (respectively $e_1 \in F_\Sigma$), and $f' \in \mathcal{T}_\Sigma$ and $U \in \mathbf{Pom}_\Sigma^+$ are such that $e_1 \xrightarrow{U}_\Sigma f'$ (respectively $e_2 \xrightarrow{U}_\Sigma f'$), then there exists an $f \in \mathcal{T}_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ with $f' \lesssim f$.

Proof. For the first claim, choose $f = \gamma_\Sigma(e_1 \parallel e_2, \llbracket e_1, e_2 \rrbracket)$. We then immediately find that $e_1 \parallel e_2 \xrightarrow{U \parallel V}_\Sigma f$ with $1 \lesssim f$.

For the second claim, the proof proceeds by induction on U . In the base, $U = a \in \Sigma$ and $f_1 = \delta_\Sigma(e_1, a)$. Choose $f = \delta_\Sigma(e_1 \parallel e_2, a)$. It is then easy to see that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \lesssim f$.

In the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , there exists a g_1 such that $e_1 \xrightarrow{V}_\Sigma g_1$ and $g_1 \xrightarrow{W}_\Sigma f_1$. By induction, we find $g \in \mathcal{T}_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{V}_\Sigma g$ with $g_1 \lesssim g$. By Lemma 4.13, we find $f \in \mathcal{T}_\Sigma$ such that $g \xrightarrow{W}_\Sigma f$ and $f_1 \lesssim f$. In total, we have that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \lesssim f$.
- If $U = V \parallel W$ with V and W smaller than U , there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{V}_\Sigma g'$ and $h \xrightarrow{W}_\Sigma h'$, and $f_1 = \gamma_\Sigma(e_1, \llbracket g, h \rrbracket)$. We choose $f = \gamma_\Sigma(e_1 \parallel e_2, \llbracket g, h \rrbracket)$. It is now easy to see that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \lesssim f$. $\square$

Lastly, we present a trace construction lemma to obtain traces originating in expressions of the form e^* . The idea here is that, given a finite number of traces that originate in e , where all (but possibly one) have an accepting state as their target, we can construct a trace originating in e^* , with a concatenation of the labels of the input traces as its label.

Lemma 4.16. *Let $e, f_1, f_2, \dots, f_n \in \mathcal{T}_\Sigma$ (with $n > 0$) be such that $f_1, f_2, \dots, f_{n-1} \in F_\Sigma$. Also, let $U, U_1, U_2, \dots, U_n \in \mathbf{Pom}_\Sigma^+$ be such that $U = U_1 \cdot U_2 \cdots U_n$. If for all $i \leq n$ it holds that $e \xrightarrow{U_i}_\Sigma f_i$, then there exists an $f \in \mathcal{T}_\Sigma$ such that $e^* \xrightarrow{U}_\Sigma f$, with $f_n \cdot e^* \lesssim f$.*

Proof. First, we show that the claim holds for $n = 1$, i.e., if $e, f_1 \in \mathcal{T}_\Sigma$ and $U \in \mathbf{Pom}_\Sigma^+$ are such that $e \xrightarrow{U}_\Sigma f_1$, then there exists an $f \in \mathcal{T}_\Sigma$ such that $e^* \xrightarrow{U}_\Sigma f$, with $f_1 \cdot e^* \lesssim f$. The proof proceeds by induction on U . In the base, $U = a \in \Sigma$ and $f_1 = \delta_\Sigma(e, a)$. We choose $f = \delta_\Sigma(e^*, a) = \delta_\Sigma(e, a) \cdot e^* = f_1 \cdot e^*$ to find that $e^* \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e^* \lesssim f$.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , there exists a $g_1 \in \mathcal{T}_\Sigma$ such that $e \xrightarrow{V}_\Sigma g_1$ and $g_1 \xrightarrow{W}_\Sigma f_1$. By induction, we find $g \in \mathcal{T}_\Sigma$ such that $e^* \xrightarrow{V}_\Sigma g$ with $g_1 \cdot e^* \lesssim g$. By Lemma 4.14, we find $h \in \mathcal{T}_\Sigma$ such that $g_1 \cdot e^* \xrightarrow{W}_\Sigma h$, with $f_1 \cdot e^* \lesssim h$. By Lemma 4.13, we find f such that $g \xrightarrow{W}_\Sigma f$ with $h \lesssim f$. In total, we have that $e^* \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e^* \lesssim f$.
- If $U = V \parallel W$ with V and W smaller than U , there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{V}_\Sigma g'$, and $h \xrightarrow{W}_\Sigma h'$, and $f_1 = \gamma_\Sigma(e, \llbracket g, h \rrbracket)$. In this case, we choose $f = \gamma_\Sigma(e^*, \llbracket g, h \rrbracket)$ to find that $e^* \xrightarrow{U}_\Sigma f$ with $f_1 \cdot e^* \lesssim f$.

We now inductively extend the claim to all $n > 0$, using the proof above as our base. In the inductive step, we assume the claim holds for $n - 1$, and try to prove it for n . Let $U' = U_1 \cdot U_2 \cdots U_{n-1}$. By induction, we obtain $g \in \mathcal{T}_\Sigma$ such that $e^* \xrightarrow{U'}_\Sigma g$ with $f_{n-1} \cdot e^* \lesssim g$. Furthermore, by the previous observation, we find $h \in \mathcal{T}_\Sigma$ such that $e^* \xrightarrow{U_n}_\Sigma h$ with $f_n \cdot e^* \lesssim h$. By Lemma 4.14 and the fact that $f_{n-1} \in F_\Sigma$, we find $d \in \mathcal{T}_\Sigma$ such that $f_{n-1} \cdot e^* \xrightarrow{U_n}_\Sigma d$ with $h \lesssim d$. By the fact that $f_{n-1} \cdot e^* \lesssim g$ and $f_{n-1} \cdot e^* \xrightarrow{U_n}_\Sigma d$, we find $f \in \mathcal{T}_\Sigma$ such that $g \xrightarrow{U_n}_\Sigma f$ with $d \lesssim f$. In total, we have that $e^* \xrightarrow{U}_\Sigma f$ with $f_n \cdot e^* \lesssim h \lesssim d \lesssim f$. $\square$

4.4 Soundness for the syntactic PA

With trace deconstruction and construction lemmas in our toolbox, we are ready to show that the syntactic PA indeed captures series-rational languages.

First, note that L_Σ can be seen as a function from $\mathcal{T}_\Sigma$ to $\mathbf{Pom}_\Sigma$, like $\llbracket - \rrbracket$. To establish equality between L_Σ and $\llbracket - \rrbracket$, we first show that L_Σ enjoys the same homomorphic equalities as those in the definition of the semantic map, i.e., that $L_\Sigma(e)$ can be expressed in terms of L_Σ applied to subexpressions of e . The proofs of the equalities below follow a similar pattern: for the inclusion from left to right we use trace deconstruction lemmas to obtain traces for the component expressions, while for the inclusion from right to left we use trace construction lemmas to build traces for the composed expressions given the traces of the component expressions. We treat the case for the empty pomset separately almost everywhere.

Lemma 4.17. *Let $e_1, e_2 \in \mathcal{T}_\Sigma$, and $a \in \Sigma$. The following equalities hold:*

$$L_\Sigma(0) = \emptyset \tag{1}$$

$$L_\Sigma(1) = \{1\} \tag{2}$$

$$L_\Sigma(a) = \{a\} \tag{3}$$

$$L_\Sigma(e_1 + e_2) = L_\Sigma(e_1) \cup L_\Sigma(e_2) \tag{4}$$

$$L_\Sigma(e_1 \cdot e_2) = L_\Sigma(e_1) \cdot L_\Sigma(e_2) \tag{5}$$

$$L_\Sigma(e_1 \parallel e_2) = L_\Sigma(e_1) \parallel L_\Sigma(e_2) \tag{6}$$

$$L_\Sigma(e_1^*) = L_\Sigma(e_1)^* \tag{7}$$

Proof. For (1), suppose that $U \in L_\Sigma(0)$. Then there exists an $f \in F_\Sigma$ such that $0 \xrightarrow{U}_{\gg\Sigma} f$. Since $0 \notin F_\Sigma$, we know that $f \neq 0$, and so $0 \xrightarrow{U}_\Sigma f$ must hold. But then, by Lemma 4.7 we know that $f = 0$, which is a contradiction. We conclude that $L_\Sigma(0) = \emptyset$.

For (2), suppose that $U \in L_\Sigma(1)$. Then there exists an $f \in F_\Sigma$ such that $1 \xrightarrow{U}_{\gg\Sigma} f$. Then either $f = 1$ and $U = 1$, or $U \in \mathbf{Pom}_\Sigma^+$ and $1 \xrightarrow{U}_\Sigma f$. The former case is possible (since $1 \in F_\Sigma$), but in the latter case we find that $f = 0$ by Lemma 4.7, and so $f \notin F_\Sigma$ — a contradiction. In conclusion, we find that $U = 1$, and thus $U \in \{1\}$. The other inclusion is easy: simply observe that $1 \xrightarrow{1}_{\gg\Sigma} 1$ by definition of $\rightarrow_\Sigma$, implying that $1 \in L_\Sigma(1)$.

For (3), suppose that $U \in L_\Sigma(a)$. Then there exists an $f \in F_\Sigma$ such that $a \xrightarrow{U}_{\gg\Sigma} f$. Then either $f = a$ and $U = 1$, or $U \in \mathbf{Pom}_\Sigma^+$ and $a \xrightarrow{U}_\Sigma f$. We can rule out the former case, as it implies that $f = a \in F_\Sigma$ — a contradiction. In the latter case, we find by Lemma 4.7 that either $U = a$ and $f = 1$, or $f = 0$. Since the latter again contradicts that $f \in F_\Sigma$, we find that $U = a$; we thus conclude that $U \in \{a\}$. The other inclusion is easy: simply observe that $a \xrightarrow{a}_\Sigma 1$ by definition of $\rightarrow_\Sigma$, and thus $a \xrightarrow{a}_{\gg\Sigma} 1$, implying that $a \in L_\Sigma(a)$.

For (4), suppose that $U \in L_\Sigma(e_1 + e_2)$. Then there exists an $f \in F_\Sigma$ such that $e_1 + e_2 \xrightarrow{U}_{\gg\Sigma} f$. If $f = e_1 + e_2$ and $U = 1$, then either $e_1 \in F_\Sigma$ or $e_2 \in F_\Sigma$. In either case, it follows quite easily that $U \in L_\Sigma(e_1) \cup L_\Sigma(e_2)$. Otherwise, $e_1 + e_2 \xrightarrow{U}_\Sigma f$. But then, by Lemma 4.8, we know that $f = f_1 + f_2$, such that $e_1 \xrightarrow{U}_\Sigma f_1$ and $e_2 \xrightarrow{U}_\Sigma f_2$. Since $f \in F_\Sigma$, either $f_1 \in F_\Sigma$ or $f_2 \in F_\Sigma$, and therefore $U \in L_\Sigma(e_1)$ or $U \in L_\Sigma(e_2)$; either way, $U \in L_\Sigma(e_1) \cup L_\Sigma(e_2)$.

To prove the other inclusion, suppose that $U \in L_\Sigma(e_1) \cup L_\Sigma(e_2)$. If $U \in L_\Sigma(e_1)$, then there exists an $f_1 \in F_\Sigma$ such that $e_1 \xrightarrow{U}_{\gg\Sigma} f_1$. If $U = 1$ and $f_1 = e_1$, then $e_1 \in F_\Sigma$ and therefore $e_1 + e_2 \in F_\Sigma$; it follows that $U \in L_\Sigma(e_1 + e_2)$. Otherwise, $e_1 \xrightarrow{U}_\Sigma f_1$, and so by Lemma 4.13 and the fact that $e_1 \lesssim e_1 + e_2$ we find $f \in \mathcal{T}_\Sigma$ such that $e_1 + e_2 \xrightarrow{U}_\Sigma f$ with $f_1 \lesssim f$. By Lemma 4.13, $f \in F_\Sigma$, and so it follows that $U \in L_\Sigma(e_1 + e_2)$. The case where $U \in L_\Sigma(e_2)$ is similar.

For (5), suppose that $U \in L_\Sigma(e_1 \cdot e_2)$. Then there exists an $f \in F_\Sigma$ such that $e_1 \cdot e_2 \xrightarrow{U}_{\gg\Sigma} f$. If $U = 1$ and $f = e_1 \cdot e_2$, then $e_1 \cdot e_2 \in F_\Sigma$, and thus $e_1, e_2 \in F_\Sigma$. Therefore $1 \in L_\Sigma(e_1)$ and $1 \in L_\Sigma(e_2)$, implying that $1 = 1 \cdot 1 \in L_\Sigma(e_1) \cdot L_\Sigma(e_2)$. Otherwise, $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$. Then, by Lemma 4.9, there exists an $f' \in \mathcal{T}_\Sigma$

and a finite set I , as well as I -indexed families of terms $(f'_i)_{i \in I} \in F_\Sigma$ and $(f_i)_{i \in I} \in \mathcal{T}_\Sigma$ and of pomsets $(U'_i)_{i \in I} \in \text{Pom}_\Sigma$ and $(U_i)_{i \in I} \in \mathcal{T}_\Sigma$, such that:

- $f \simeq f' \cdot e_2 + \sum_{i \in I} f_i, e_1 \xrightarrow{U}_\Sigma f'$
- for all $i \in I$, $e_1 \xrightarrow{U'_i}_\Sigma f'_i, e_2 \xrightarrow{U_i}_\Sigma f_i$ and $U = U'_i \cdot U_i$.

Since $f \in F_\Sigma$, also $f' \cdot e_2 + \sum_{i \in I} f_i \in F_\Sigma$ by Lemma 4.4. This means that either $f' \cdot e_2 \in F_\Sigma$, or $f_i \in F_\Sigma$ for some $i \in I$. In the former case, $f', e_2 \in F_\Sigma$, and therefore $U \in L_\Sigma(e_1)$ and $1 \in L_\Sigma(e_2)$, thus $U = U \cdot 1 \in L_\Sigma(e_1) \cdot L_\Sigma(e_2)$. In the latter case, $U'_i \in L_\Sigma(e_1)$ and $U_i \in L_\Sigma(e_2)$, thus $U = U'_i \cdot U_i \in L_\Sigma(e_1) \cdot L_\Sigma(e_2)$.

To prove the other inclusion, suppose that $U \in L_\Sigma(e_1) \cdot L_\Sigma(e_2)$. Then $U = U_1 \cdot U_2$, with $U_1 \in L_\Sigma(e_1)$ and $U_2 \in L_\Sigma(e_2)$. There are four cases to consider.

- If $U_1 = 1$ and $U_2 = 1$, then $e_1, e_2 \in F_\Sigma$, and therefore $e_1 \cdot e_2 \in F_\Sigma$. Thus $1 \in L_\Sigma(e_1 \cdot e_2)$.
- If $U_1 = 1$ and $U_2 \neq 1$, then $e_1 \in F_\Sigma$ and $e_2 \xrightarrow{U_2}_\Sigma f_2$ for some $f_2 \in F_\Sigma$. Then, by Lemma 4.14, $e_1 \cdot e_2 \xrightarrow{U_2}_\Sigma f$ for some $f \in \mathcal{T}_\Sigma$ with $f_2 \lesssim f$. By Lemma 4.13, $f \in F_\Sigma$, and thus $U = U_2 \in L_\Sigma(e_1 \cdot e_2)$.
- If $U_1 \neq 1$ and $U_2 = 1$, then $e_1 \xrightarrow{U_1}_\Sigma f_1$ for some $f_1 \in F_\Sigma$, and $e_2 \in F_\Sigma$. Then, by Lemma 4.14, $e_1 \cdot e_2 \xrightarrow{U_1}_\Sigma f$ for some $f \in \mathcal{T}_\Sigma$ with $f_1 \cdot e_2 \lesssim f$. Since $f_1, e_2 \in F_\Sigma$, also $f_1 \cdot e_2 \in F_\Sigma$, and thus by Lemma 4.13, $f \in F_\Sigma$. Therefore $U = U_1 \in L_\Sigma(e_1 \cdot e_2)$.
- If $U_1 \neq 1$ and $U_2 \neq 1$, then $e_1 \xrightarrow{U_1}_\Sigma f_1$ and $e_2 \xrightarrow{U_2}_\Sigma f_2$ for some $f_1, f_2 \in F_\Sigma$. Then, by Lemma 4.14, we find that $e_1 \cdot e_2 \xrightarrow{U}_\Sigma f$ with $f_2 \lesssim f$. By Lemma 4.13, $f \in F_\Sigma$, and thus $U \in L_\Sigma(e_1 \cdot e_2)$.

For (6), suppose that $U \in L_\Sigma(e_1 \parallel e_2)$. Then there exists an $f \in F_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$. If $f = e_1 \parallel e_2$ and $U = 1$, then $e_1, e_2 \in F_\Sigma$, and thus $1 \in L_\Sigma(e_1)$ and $1 \in L_\Sigma(e_2)$. But then $1 = 1 \parallel 1 \in L_\Sigma(e_1) \parallel L_\Sigma(e_2)$. Otherwise, $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ for some $f \in F_\Sigma$. By Lemma 4.10, we obtain $f_1, f_2, f_3 \in F_\Sigma$ such that

- $f \simeq f_1 + f_2 + f_3$
- either $f_1 = 0$, or $e_2 \in F_\Sigma$ and $e_1 \xrightarrow{U}_\Sigma f_1$
- either $f_2 = 0$, or $e_1 \in F_\Sigma$ and $e_2 \xrightarrow{U}_\Sigma f_2$
- either $f_3 = 0$, or $f_3 = 1$ and there exist $e'_1, e'_2 \in \mathcal{T}_\Sigma$ and $f'_1, f'_2 \in F_\Sigma$ and $U_1, U_2 \in \text{Pom}_\Sigma^+$ such that $e_1 \simeq e'_1$ and $e_2 \simeq e'_2$ and $U = U_1 \parallel U_2$ and $e'_1 \xrightarrow{U_1}_\Sigma f'_1$ and $e'_2 \xrightarrow{U_2}_\Sigma f'_2$.

By Lemma 4.13, $f_1 + f_2 + f_3 \in F_\Sigma$, and thus $f_1 \in F_\Sigma$ or $f_2 \in F_\Sigma$ or $f_3 \in F_\Sigma$. In the first case, $U \in L_\Sigma(e_1)$ and $1 \in L_\Sigma(e_2)$, and therefore $U = U \parallel 1 \in L_\Sigma(e_1) \parallel L_\Sigma(e_2)$. In the second case, we similarly find that $U \in L_\Sigma(e_1) \parallel L_\Sigma(e_2)$. In the last case, we find by Lemma 4.6 and Lemma 4.4 $f'_1, f'_2 \in F_\Sigma$ such that $e_1 \xrightarrow{U_1}_\Sigma f'_1$ and $e_2 \xrightarrow{U_2}_\Sigma f'_2$, and thus we have that $U = U_1 \parallel U_2$ such that $U_1 \in L_\Sigma(e_1)$ and $U_2 \in L_\Sigma(e_2)$, therefore $U \in L_\Sigma(e_1) \parallel L_\Sigma(e_2)$.

To prove the other inclusion, suppose that $U \in L_\Sigma(e_1) \parallel L_\Sigma(e_2)$. Then $U = U_1 \parallel U_2$. There are four cases to consider.

- If $U_1 = 1$ and $U_2 = 1$, then $e_1, e_2 \in F_\Sigma$, thus $e_1 \parallel e_2 \in F_\Sigma$. But then $U = 1 \in L_\Sigma(e_1 \parallel e_2)$.
- If $U_1 = 1$ and $U_2 \neq 1$, then $e_2 \xrightarrow{U_2}_\Sigma f_2$ for some $f_2 \in F_\Sigma$. By Lemma 4.15, we find an $f \in \mathcal{T}_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ with $f_2 \lesssim f$. But then, by Lemma 4.13, $f \in F_\Sigma$, and thus $U \in L_\Sigma(e_1 \parallel e_2)$.
- If $U_1 \neq 1$ and $U_2 = 1$, then we find that $U \in L_\Sigma(e_1 \parallel e_2)$ by an argument similar to the above case.
- If $U_1 \neq 1$ and $U_2 \neq 1$, then $e_1 \xrightarrow{U_1}_\Sigma f_1$ and $e_2 \xrightarrow{U_2}_\Sigma f_2$ for some $f_1, f_2 \in F_\Sigma$. By Lemma 4.15, we find an $f \in \mathcal{T}_\Sigma$ such that $e_1 \parallel e_2 \xrightarrow{U}_\Sigma f$ with $1 \lesssim f$. But then, by Lemma 4.13, $f \in F_\Sigma$, and thus $U \in L_\Sigma(e_1 \parallel e_2)$.

For (7), suppose that $U \in L_\Sigma(e_1^*)$. Then there exists an $f \in F_\Sigma$ such that $e_1^* \xrightarrow{U}_\Sigma f$. If $f = e_1^*$ and $U = 1$, then $1 \in L_\Sigma(e_1^*)$. Otherwise, $e_1 \xrightarrow{U}_\Sigma f$. By Lemma 4.11, we find a finite set I and an I -indexed family of finite sets $(J_i)_{i \in I}$ as well as I -indexed families of terms $(f_i)_{i \in I} \in \mathcal{T}_\Sigma$ and of pomsets $(U_i)_{i \in I} \in \text{Pom}_\Sigma^+$, and for all $i \in I$ also J_i -indexed families of terms $(f_{i,j})_{j \in J_i}$ and of pomsets $(U_{i,j})_{j \in J_i}$, such that $f \simeq \sum_{i \in I} f_i \cdot e^*$, and for all $i \in I$

- $e \xrightarrow{U_i}_{\Sigma} f_i$
- for all $j \in J_i$ we have that $e \xrightarrow{U_{i,j}}_{\Sigma} f_{i,j}$
- $U = U'_i \cdot U_i$, where U'_i is some concatenation of all $U_{i,j}$ for all $j \in J_i$.

By Lemma 4.13, we know that $\sum_{i \in I} f_i \cdot e^* \in F_{\Sigma}$, and thus $f_i \cdot e^* \in F_{\Sigma}$ for some $i \in I$, meaning in particular that $f_i \in F_{\Sigma}$ for this $i \in I$. Now $U_{i,j} \in L_{\Sigma}(e)$ for all $j \in J$, and thus $U'_i \in L_{\Sigma}(e)^*$. Since also $U_i \in L_{\Sigma}(e)$, we find that $U = U'_i \cdot U_i \in L_{\Sigma}(e)^* \cdot L_{\Sigma}(e) \subseteq L_{\Sigma}(e)^*$.

To prove the other inclusion, suppose that $U \in L_{\Sigma}(e)^*$. Then $U \in L_{\Sigma}(e)^n$, i.e., $U = U_1 \cdot U_2 \cdots U_n$, such that for all $1 \leq i \leq n$ we have $U_i \in L_{\Sigma}(e)$. Assume (without loss of generality) that $U_i \neq 1$ for all $1 \leq i \leq n$. If $n = 0$, then $U = 1$ and we find that $U \in L_{\Sigma}(e^*)$ immediately. Otherwise, there exist $f_1, f_2, \dots, f_n \in F_{\Sigma}$ such that $e \xrightarrow{U_i}_{\Sigma} f_i$. By Lemma 4.16 we find $f \in \mathcal{T}_{\Sigma}$ such that $e^* \xrightarrow{U}_{\Sigma} f$ with $f_n \lesssim f$. But then $f \in F_{\Sigma}$ by Lemma 4.13, and thus $U \in L_{\Sigma}(e^*)$. $\square$

It is now easy to establish that the Brzozowski construction for the syntactic PA is sound with respect to the denotational semantics of series-rational expressions.

Theorem 4.18. *Let $e \in \mathcal{T}_{\Sigma}$. Then $L_{\Sigma}(e) = \llbracket e \rrbracket$.*

Proof. The proof proceeds by induction on e . In the base, $e = 0$, $e = 1$ or $e = a$ for some $a \in \Sigma$. In all cases, $L_{\Sigma}(e) = \llbracket e \rrbracket$ by Lemma 4.17. For the inductive step, there are four cases to consider: either $e = e_1 + e_2$, $e = e_1 \cdot e_2$, $e = e_1 \parallel e_2$ or $e = e_1^*$. In all cases, the claim follows from the induction hypothesis and the definition of $\llbracket - \rrbracket$, combined with Lemma 4.17. $\square$

4.5 Bounding the syntactic PA

Ideally, we would like to obtain a single PA with finitely many states that recognizes $\llbracket e \rrbracket$ for a given $e \in \mathcal{T}_{\Sigma}$. Unfortunately, the syntactic PA is not bounded, and thus Theorem 3.12 does not apply. For instance, the requirement that $\rho_{\Sigma}(e)$ be finite for $e \in \mathcal{T}_{\Sigma}$ fails; consider the family of distinct terms $(e_n)_{n \in \mathbb{N}}$ defined by $e_0 = 1 \cdot a^*$ and $e_{n+1} = 0 \cdot a^* + e_n$ for $n \in \mathbb{N}$; it is not hard to show that $e_n \in \rho_{\Sigma}(a^*)$ for $n \in \mathbb{N}$, and thus conclude that $\rho_{\Sigma}(a^*)$ is infinite. We remedy this problem by quotienting the state space of the syntactic PA by congruence.

In what follows, we write $[e]$ for the congruence class of $e \in \mathcal{T}_{\Sigma}$ modulo $\simeq$, i.e., the set of all $e' \in \mathcal{T}_{\Sigma}$ such that $e \simeq e'$. We furthermore write $\mathcal{Q}_{\Sigma}$ for the set of all congruence classes of expressions in $\mathcal{T}_{\Sigma}$. We now leverage Lemma 4.5 to define a transition structure on $\mathcal{Q}_{\Sigma}$.

Definition 4.19. We define $\delta_{\simeq} : \mathcal{Q}_{\Sigma} \times \Sigma \rightarrow \mathcal{Q}_{\Sigma}$ and $\gamma_{\simeq} : \mathcal{Q}_{\Sigma} \times \left(\frac{\mathcal{Q}_{\Sigma}}{2}\right) \rightarrow \mathcal{Q}_{\Sigma}$ as

$$\delta_{\simeq}([e], a) = [\delta_{\Sigma}(e, a)] \quad \gamma_{\simeq}([e], \{[f], [g]\}) = \begin{cases} [0] & f \simeq 0 \text{ or } g \simeq 0 \\ [\gamma_{\Sigma}(e, \{g, h\})] & \text{otherwise} \end{cases}$$

Furthermore, the set $F_{\simeq}$ is defined to be $\{[e] : e \in F_{\Sigma}\}$. The *quotiented syntactic PA* is the PA $A_{\simeq} = \langle \mathcal{Q}_{\Sigma}, \delta_{\simeq}, \gamma_{\simeq}, F_{\simeq} \rangle$.

Lemma 4.20. *$\delta_{\simeq}$ and $\gamma_{\simeq}$, as well as $F_{\simeq}$, are well-defined.*

Proof. The proof for $\delta_{\simeq}$ and $\gamma_{\simeq}$ follows from Lemma 4.5. The proof for $F_{\simeq}$ follows from Lemma 4.4. $\square$

Just like for the syntactic PA, we abbreviate subscripts, for example by writing $\rightarrow_{\simeq}$ rather than $\rightarrow_{A_{\simeq}}$, and $L_{\simeq}$ rather than $L_{A_{\simeq}}$. Of course, we also want the quotiented syntactic PA to accept the same languages as the syntactic PA. To that end, we show that the trace relations of the syntactic PA and the quotiented syntactic PA correspond.

Lemma 4.21. *Let $e, f \in \mathcal{T}_{\Sigma}$ and $U \in \text{Pom}_{\Sigma}^+$. If $e \xrightarrow{U}_{\Sigma} f$, then $[e] \xrightarrow{U}_{\simeq} [f]$. If $[e] \xrightarrow{U}_{\simeq} [f]$, then there exists an $f' \in \mathcal{T}_{\Sigma}$ with $f \simeq f'$ and $e \xrightarrow{U}_{\Sigma} f'$.*

Proof. Refer to Appendix A. $\square$

Theorem 4.22. *Let $e \in \mathcal{T}_{\Sigma}$. Then $L_{\Sigma}(e) = L_{\simeq}([e])$.*

Proof. By definition, $e \in F_{\Sigma}$ if and only if $[e] \in F_{\simeq}$. It then follows that $1 \in L_{\Sigma}(e)$ if and only if $1 \in L_{\simeq}([e])$. The remaining cases are covered by Lemma 4.21. $\square$

Corollary 4.23. *The state $[0]$ is the only state in the quotiented syntactic PA with an empty language.*

Proof. Let $[e] \in \mathcal{Q}_\Sigma$ be a state such that $L_\simeq([e]) = \emptyset$. Then, by Theorem 4.22, we know that $L_\Sigma(e) = \emptyset$, and by Theorem 4.18, $\llbracket e \rrbracket = \emptyset$. But then $e \simeq 0$ by Lemma 2.11, and thus $[e] = [0]$. $\square$

To show that the quotiented syntactic PA is bounded, we construct a strict order $\prec$ on $\mathcal{Q}_\Sigma$ that satisfies the rules of Definition 3.8.

Definition 4.24. Let $e \in \mathcal{T}_\Sigma$. The *parallel depth* of e , denoted $\langle\!\langle e \rangle\!\rangle$, is defined to be 0 when $e \simeq 0$, and otherwise:

$$\begin{aligned} \langle\!\langle 1 \rangle\!\rangle &= 0 & \langle\!\langle a \rangle\!\rangle &= 1 & \langle\!\langle e_1 + e_2 \rangle\!\rangle &= \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) \\ \langle\!\langle e_1 \cdot e_2 \rangle\!\rangle &= \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) & \langle\!\langle e_1 \parallel e_2 \rangle\!\rangle &= \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) + 1 & \langle\!\langle e_1^* \rangle\!\rangle &= \langle\!\langle e_1 \rangle\!\rangle \end{aligned}$$

It is easy to show that the parallel depth of an expression is also well-defined on the congruence classes of $\simeq$. This allows us to define the desired strict order on $\mathcal{Q}_\Sigma$.

Lemma 4.25. *Let $e, e' \in \mathcal{T}_\Sigma$ be such that $e \simeq e'$. Then $\langle\!\langle e \rangle\!\rangle = \langle\!\langle e' \rangle\!\rangle$.*

Proof. Refer to Appendix A. $\square$

Definition 4.26. The relation $\prec \subseteq \mathcal{Q}_\Sigma \times \mathcal{Q}_\Sigma$ is such that $[e] \prec [f]$ if and only if $\langle\!\langle e \rangle\!\rangle < \langle\!\langle f \rangle\!\rangle$.

As it turns out, $\prec$ is the desired fork hierarchy.

Lemma 4.27. *The quotiented syntactic PA is fork-acyclic, with fork hierarchy $\prec$.*

Proof. We need to show that $\prec$ satisfies the conditions of Definition 3.8.

For the first rule, we begin by showing that if $g, h, e \in \mathcal{T}_\Sigma$ such that $e, g, h \neq 0$ and $\gamma_\Sigma(e, \{g, h\}) \neq 0$, then $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e \rangle\!\rangle$. The proof proceeds by induction on e . In the base, $e = 1$ or $e = a$; in both cases, $\gamma_\Sigma(e, \{g, h\}) = 0$ and so the claim holds vacuously. For the inductive step, there are four cases to consider.

- If $e = e_1 + e_2$, then $\gamma_\Sigma(e, \{g, h\}) = \gamma_\Sigma(e_1, \{g, h\}) + \gamma_\Sigma(e_2, \{g, h\}) \neq 0$, thus $\gamma_\Sigma(e_1, \{g, h\}) \neq 0$ or $\gamma_\Sigma(e_2, \{g, h\}) \neq 0$. Therefore, by induction $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e_1 \rangle\!\rangle$ or $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e_2 \rangle\!\rangle$, and thus $\langle\!\langle g \rangle\!\rangle < \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) = \langle\!\langle e \rangle\!\rangle$.
- If $e = e_1 \cdot e_2$, then $\gamma_\Sigma(e, \{g, h\}) = \gamma_\Sigma(e_1, \{g, h\}) \cdot e_2 + e_1 \star \gamma_\Sigma(e_2, \{g, h\}) \neq 0$, thus $\gamma_\Sigma(e_1, \{g, h\}) \neq 0$ or $\gamma_\Sigma(e_2, \{g, h\}) \neq 0$. Therefore, by induction $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e_1 \rangle\!\rangle$ or $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e_2 \rangle\!\rangle$, and thus $\langle\!\langle g \rangle\!\rangle < \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) = \langle\!\langle e \rangle\!\rangle$.
- If $e = e_1 \parallel e_2$, then $\gamma_\Sigma(e, \{g, h\}) = [\{g, h\} = \{e_1, e_2\}] + e_2 \star \gamma_\Sigma(e_1, \{g, h\}) + e_1 \star \gamma_\Sigma(e_2, \{g, h\}) \neq 0$. If $[\{g, h\} = \{e_1, e_2\}] \neq 0$, we know that $\{g, h\} = \{e_1, e_2\}$. Assume without loss of generality that $e_1 = g$; then $\langle\!\langle g \rangle\!\rangle = \langle\!\langle e_1 \rangle\!\rangle < \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) + 1 = \langle\!\langle e \rangle\!\rangle$. If $\gamma_\Sigma(e_1, \{g, h\}) \neq 0$, then by induction $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e_1 \rangle\!\rangle < \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) + 1$. The case where $\gamma_\Sigma(e_2, \{g, h\}) \neq 0$ is similar.
- If $e = e_1^*$, then $\gamma_\Sigma(e, \{g, h\}) = \gamma_\Sigma(e_1, \{g, h\}) \cdot e_1^* \neq 0$ and therefore $\gamma_\Sigma(e_1, \{g, h\}) \neq 0$. But then $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e_1 \rangle\!\rangle = \langle\!\langle e \rangle\!\rangle$ by induction.

To fully validate the first rule, suppose that $\{[g], [h]\} \in \pi_\simeq([e])$. Then $e, g, h \neq 0$ and $\gamma_\simeq([e], \{[g], [h]\}) \neq [0]$ and thus $\gamma_\Sigma(e, \{g, h\}) \neq 0$. But then we find that $\langle\!\langle g \rangle\!\rangle < \langle\!\langle e \rangle\!\rangle$ by the above, and therefore $[g] \prec [e]$.

For the second rule, we first show that if $a \in \Sigma$ and $e \in \mathcal{T}_\Sigma$, then $\langle\!\langle \delta_\Sigma(e, a) \rangle\!\rangle \leq \langle\!\langle e \rangle\!\rangle$. If $e \simeq 0$, then $\langle\!\langle \delta_\Sigma(e, a) \rangle\!\rangle = \langle\!\langle \delta_\Sigma(0, a) \rangle\!\rangle = \langle\!\langle 0 \rangle\!\rangle = 0 \leq \langle\!\langle 0 \rangle\!\rangle = \langle\!\langle e \rangle\!\rangle$. The proof of the remaining cases proceeds by induction on e . In the base, $e = 1$ or $e = a$. In both cases, $\langle\!\langle \delta_\Sigma(e, a) \rangle\!\rangle = 0 \leq \langle\!\langle e \rangle\!\rangle$.

For the inductive step, there are four cases to consider.

- If $e = e_1 + e_2$, then we can derive

$$\langle\!\langle \delta_\Sigma(e, a) \rangle\!\rangle = \langle\!\langle \delta_\Sigma(e_1, a) + \delta_\Sigma(e_2, a) \rangle\!\rangle = \max(\langle\!\langle \delta_\Sigma(e_1, a) \rangle\!\rangle, \langle\!\langle \delta_\Sigma(e_2, a) \rangle\!\rangle) \leq \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) = \langle\!\langle e \rangle\!\rangle$$

- If $e = e_1 \cdot e_2$, then we can derive

$$\begin{aligned} \langle\!\langle \delta_\Sigma(e, a) \rangle\!\rangle &= \langle\!\langle \delta_\Sigma(e_1, a) \cdot e_2 + e_1 \star \delta_\Sigma(e_2, a) \rangle\!\rangle \\ &\leq \max(\langle\!\langle \delta_\Sigma(e_1, a) \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle, \langle\!\langle \delta_\Sigma(e_2, a) \rangle\!\rangle) \\ &\leq \max(\langle\!\langle e_1 \rangle\!\rangle, \langle\!\langle e_2 \rangle\!\rangle) = \langle\!\langle e \rangle\!\rangle \end{aligned}$$

- If $e = e_1 \parallel e_2$, then we can derive

$$\begin{aligned} \langle \delta_\Sigma(e, a) \rangle &= \langle e_2 \star \delta_\Sigma(e_1, a) + e_1 \star \delta_\Sigma(e_2, a) \rangle \\ &\leq \max(\langle \delta_\Sigma(e_1, a) \rangle, \langle \delta_\Sigma(e_2, a) \rangle) \\ &\leq \max(\langle e_1 \rangle, \langle e_2 \rangle) = \langle e_1 \parallel e_2 \rangle \end{aligned}$$

- If $e = e_1^*$, then we can derive

$$\langle \delta_\Sigma(e, a) \rangle = \langle \delta_\Sigma(e_1, a) \cdot e_1^* \rangle \leq \max(\langle \delta_\Sigma(e_1, a) \rangle, \langle e_1^* \rangle) \leq \langle e \rangle$$

Thus, suppose that $[f] \prec \delta_\Sigma([e], a) = [\delta_\Sigma(e, a)]$. Then $[f] \prec [e]$, since we can derive that $\langle f \rangle < \langle \delta_\Sigma(e, a) \rangle \leq \langle e \rangle$.

For the third rule, one also first shows that for $e \in \mathcal{T}_\Sigma$ and $\phi \in \binom{\mathcal{T}_\Sigma}{2}$ we have that $\langle \gamma_\Sigma(e, \phi) \rangle \leq \langle e \rangle$. One can then extend this to show that if $[f] \prec \gamma_\Sigma([e], \phi)$, also $[f] \prec [e]$ using the same technique as above. $\square$

We now investigate the reach and support of a state in the quotiented syntactic PA, using the deconstruction lemmas showed in Section 4.2, as well as Lemma 4.21.

Lemma 4.28. *Let $e \in \mathcal{T}_\Sigma$. Then $\rho_\Sigma([e])$ and $\pi_\Sigma([e])$ are finite.*

Proof. We first claim that $R_\Sigma(e)$ is finite up to $\simeq$, i.e., that the set $\{[f] : f \in R_\Sigma(e)\}$ is finite. The proof of this claim proceeds by induction on e . In the base, there are three cases to consider.

- If $e = 0$ and $e \xrightarrow{\cup}_\Sigma f$, then $f = 0$ by Lemma 4.7; it follows that $R_\Sigma(e)$ is finite.
- If $e = 1$ and $e \xrightarrow{\cup}_\Sigma f$, then $f = 0$ by Lemma 4.7; it follows that $R_\Sigma(e)$ is finite.
- If $e = a$ for some $a \in \Sigma$, then $f = 0$ or $f = 1$ by Lemma 4.7. It follows that $R_\Sigma(e)$ is finite.

In all cases, we find that $R_\Sigma(e)$ is finite, and therefore finite up to $\simeq$.

For the inductive step, there are four cases to consider. In each case our strategy is to write an arbitrary element of $R_\Sigma(e)$ into a sum of a number of terms. If we can then show that there are only finitely many choices for each term up to $\simeq$, it follows that there are only finitely many such sums up to $\simeq$. Note that the number of terms does not matter, since repeated terms in a sum do not make a difference with respect to $\simeq$ by virtue of the idempotence rule.

- If $e = e_1 + e_2$ and $f \in R_\Sigma(e)$, then by Lemma 4.8 there exist $f_1 \in R_\Sigma(e_1)$ and $f_2 \in R_\Sigma(e_2)$ such that $f \simeq f_1 + f_2$. Since $R_\Sigma(e_1)$ and $R_\Sigma(e_2)$ are finite up to $\simeq$ by induction, it follows that $R_\Sigma(e)$ is finite up to $\simeq$.
- If $e = e_1 \cdot e_2$ and $f \in R_\Sigma(e)$, then by Lemma 4.9 there exists an $f' \in R_\Sigma(e_1) \cup \{e_1\}$ and a finite set I and an I -indexed set of terms $(f_i)_{i \in I} \in R_\Sigma(e_2)$ such that $f \simeq f' \cdot e_2 + \sum_{i \in I} f_i$. Since $R_\Sigma(e_1) \cup \{e_1\}$ and $R_\Sigma(e_2)$ are finite up to $\simeq$ by induction, it follows that $R_\Sigma(e)$ is finite up to $\simeq$.
- If $e = e_1 \parallel e_2$ and $f \in R_\Sigma(e)$, then by Lemma 4.10 there exist $f_1 \in R_\Sigma(e_1) \cup \{0\}$, $f_2 \in R_\Sigma(e_2) \cup \{0\}$ and $f_3 \in \{0, 1\}$ such that $f \simeq f_1 + f_2 + f_3$. Since $R_\Sigma(e_1)$ and $R_\Sigma(e_2)$ are finite up to $\simeq$ by induction, it follows that $R_\Sigma(e)$ is finite up to $\simeq$.
- If $e = e_1^*$ and $f \in R_\Sigma(e)$, then by Lemma 4.11 there exists a finite set I and an I -indexed family of terms $(f_i)_{i \in I} \in R_\Sigma(e_1)$ such that $f \simeq \sum_{i \in I} f_i \cdot e^*$. Since $R_\Sigma(e_1)$ is finite up to $\simeq$ by induction, it follows that $R_\Sigma(e)$ is finite up to $\simeq$.

Since $\rho_\Sigma(e) = R_\Sigma([e]) \cup \{[e], [0]\}$ by Lemma 3.7 and Corollary 4.23, it suffices to show that $R_\Sigma([e])$ is finite. This is an immediate consequence of Lemma 4.21 and the above.

We now treat the claim about $\pi_\Sigma([e])$. First, we claim that there are finitely many $\phi \in \binom{\mathcal{T}_\Sigma}{2}$ up to $\simeq$ such that $\gamma_\Sigma(e, \phi) \neq 0$. This is shown by induction on e . In the base, $e = 0$, $e = 1$ or $e = a$; in all cases, we find that $\gamma_\Sigma(e, \phi) = 0$ for $\phi \in \binom{\mathcal{T}_\Sigma}{2}$. For the inductive step, there are four cases; in each case, we apply Lemma 4.17.

- Suppose $e = e_1 + e_2$. By induction, there are finitely many ϕ up to $\simeq$ such that $\gamma_\Sigma(e_1, \phi) \neq 0$ or $\gamma_\Sigma(e_2, \phi) \neq 0$. Thus, there are finitely many $\phi \in \binom{\mathcal{T}_\Sigma}{2}$ up to $\simeq$ such that $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) + \gamma_\Sigma(e_2, \phi) \neq 0$.

- Suppose $e = e_1 \cdot e_2$. By induction, there are finitely many ϕ up to $\simeq$ such that $\gamma_\Sigma(e_1, \phi) \neq 0$ or $\gamma_\Sigma(e_2, \phi) \neq 0$. Thus, there are finitely many $\phi \in \binom{\mathcal{T}_\Sigma}{2}$ up to $\simeq$ such that $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot e_2 + e_1 \star \gamma_\Sigma(e_2, \phi) \neq 0$.
- Suppose $e = e_1 \parallel e_2$. By induction, there are finitely many ϕ up to $\simeq$ such that $\gamma_\Sigma(e_1, \phi) \neq 0$ or $\gamma_\Sigma(e_2, \phi) \neq 0$. Furthermore, there is exactly one ϕ up to $\simeq$ such that $\phi \simeq \llbracket e_1, e_2 \rrbracket$. Thus, there are finitely many $\phi \in \binom{\mathcal{T}_\Sigma}{2}$ up to $\simeq$ such that $\gamma_\Sigma(e, \phi) = [\phi \simeq \llbracket e_1, e_2 \rrbracket] + e_2 \star \gamma_\Sigma(e_1, \phi) + e_1 \star \gamma_\Sigma(e_2, \phi) \neq 0$.
- Suppose $e = e_1^*$. By induction, there are finitely many ϕ up to $\simeq$ such that $\gamma_\Sigma(e_1, \phi) \neq 0$. Thus, there are finitely many $\phi \in \binom{\mathcal{T}_\Sigma}{2}$ up to $\simeq$ such that $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot e_1^* \neq 0$.

The proof that that $\pi_\simeq([e])$ is finite, i.e., that there are finitely many $\phi \in \binom{\mathcal{Q}_\Sigma}{2}$ such that $\gamma_\simeq([e], \phi) \neq [0]$ is now an immediate consequence of the above. $\square$

Theorem 4.29. *The quotiented syntactic PA is bounded.*

Proof. By Lemma 4.27, we know that $A_\simeq$ is fork-acyclic. Furthermore, $\prec$ is well-founded by construction: every term has finite depth. By Lemma 4.28, we know that $\rho_\simeq([e])$ and $\pi_\simeq([e])$ are finite for all $[e] \in \mathcal{Q}_\Sigma$. All requirements are now validated. $\square$

Since the quotiented syntactic PA is bounded, it restricts to a finite PA for any expression.

Corollary 4.30. *Let $e \in \mathcal{T}_\Sigma$. There exists a finite PA A_e that accepts $\llbracket e \rrbracket$.*

Proof. Consider the quotiented syntactic PA $A_\simeq$. By Theorem 4.22 and Theorem 4.18, we know that $L_\simeq([e]) = L_\Sigma(e) = \llbracket e \rrbracket$. By Theorem 4.29, $A_\simeq$ is bounded, and therefore (by Theorem 3.12) we can construct a finite and closed set Q_e such that $[e] \in Q_e$. We now choose $A_e = A_\simeq \upharpoonright_{Q_e}$ and $q = [e]$ to find that $L_{A_e}(q) = L_\simeq([e]) = \llbracket e \rrbracket$ by Lemma 3.5. $\square$

5 Automata to expressions

To associate with every state q in a bounded PA $A = \langle Q, \delta, \gamma, F \rangle$ a series-rational expression e_q such that $\llbracket e_q \rrbracket = L_A(q)$, we modify the procedure for associating a regular expression with a state in a finite automaton described in [10]. The modification consists of adding parallel terms to the expression associated with q whenever a fork in q contributes to its language, i.e., whenever $\llbracket r, s \rrbracket \in \pi_A(q)$.

In view of the special treatment of 1 in the semantics of PAs, it is convenient to first define expressions e_q^+ with the property that $\llbracket e_q^+ \rrbracket = L_A(q) - \{1\}$; then we can define e_q by $e_q = e_q^+ + [q \in F]$. The definition of e_q^+ proceeds by induction on the well-founded partial order $\prec_A$ associated with a bounded PA. That is, when defining e_q^+ we assume the existence of expressions $e_{q'}^+$ for all $q' \in Q$ such that $q' \prec_A q$.

First, however, we shall define auxiliary expressions $e_{qq'}^{Q'}$ for suitable choices of $Q' \subseteq Q$ and of $q, q' \in Q$. Intuitively, $e_{qq'}^{Q'}$ denotes the pomset language characterizing all paths from q to q' with all intermediate states in Q' ; e_q^+ can then be defined as the summation of all $e_{qq'}^{Q'}$ with $q' \in F \cap \rho_A(q)$.

Definition 5.1. Let Q' be a finite subset of Q , and assume that for all $r \in Q$ such that $r \prec_A q$ for some $q \in Q'$ there exists a series-rational expression $e_r^+ \in \mathcal{T}_\Sigma$ such that $\llbracket e_r^+ \rrbracket = L_A(r) - \{1\}$. For all $Q'' \subseteq Q'$ and $q, q' \in Q'$, we define a series-rational expression $e_{qq'}^{Q''}$ by induction on the size of Q'' , as follows:

1. If $Q'' = \emptyset$, then let $\tilde{\Sigma}$ be the set $\{a \in \Sigma : q' = \delta(q, a)\}$, and let $\tilde{Q}$ be the set $\{\phi \in \pi_A(q) : \gamma(q, \phi) = q'\}$. We define

$$e_{qq'}^{Q''} = \sum_{a \in \tilde{\Sigma}} a + \sum_{\llbracket r, s \rrbracket \in \tilde{Q}} e_r^+ \parallel e_s^+.$$

2. Otherwise, we choose a $q'' \in Q''$ and define

$$e_{qq'}^{Q''} = e_{qq'}^{Q'' - \{q''\}} + e_{qq''}^{Q'' - \{q''\}} \cdot (e_{q''q''}^{Q'' - \{q''\}})^* \cdot e_{q''q'}^{Q'' - \{q''\}}.$$

Note that e_r^+ and e_s^+ , appearing in the first clause of the definition of $e_{qq'}^{Q''}$, exist by assumption, for by fork-acyclicity we have that $r, s \prec_A q \in Q'$.

Theorem 5.2. Let Q' be a finite subset of Q and assume that for all $r \in Q$ such that $r \prec_A q$ for some $q \in Q'$ there exists a series-rational expression $e_r^+ \in \mathcal{T}_\Sigma$ with $\llbracket e_r^+ \rrbracket = L_A(q) - \{1\}$. For all $q, q' \in Q'$, for all $Q'' \subseteq Q'$, and for all $U \in \text{Pom}_\Sigma^+$, we have that $q \xrightarrow{U}_A q'$ according to some path that only visits states in Q'' if, and only if, $U \in \llbracket e_{qq'}^{Q''} \rrbracket$.

Proof. We prove the implication from left to right with induction on the size of Q'' .

Suppose that $q \xrightarrow{U}_A q'$ according to some path through A that only visits states in Q'' . Then there exist $n \geq 1$, $q_0, q_n \in Q$, $q_1, \dots, q_{n-1} \in Q''$ and $U_1, \dots, U_n \in \text{Pom}_\Sigma^+$ such that $U = U_1 \cdots U_n$ and

$$q = q_0 \xrightarrow{U_1} q_1 \xrightarrow{U_2} \cdots \xrightarrow{U_{n-1}} q_{n-1} \xrightarrow{U_n} q_n = q'$$

Furthermore, for all $1 \leq i \leq n$, either $U_i = a$ for some $a \in \Sigma$ and $q_i = \delta(q_{i-1}, a)$, or $U_i = V_i \parallel W_i$ for some $V_i, W_i \in \text{Pom}_\Sigma^+$ and there are $r_i, s_i \in Q$ and $r'_i, s'_i \in F$ such that $r_i \xrightarrow{V_i} r'_i$, $s_i \xrightarrow{W_i} s'_i$ and $q_i = \gamma(q_{i-1}, \{r_i, s_i\})$. We distinguish cases according to whether $n = 1$ or $n > 1$.

- If $n = 1$, then we distinguish cases according to whether $U_1 = a$ or $U_1 = V_1 \parallel W_1$. In the first case, we have that $q' = \delta(q, a)$, so $U = U_1 = a \in \llbracket e_{qq'}^\emptyset \rrbracket \subseteq \llbracket e_{qq'}^{Q''} \rrbracket$. In the second case, we have that there exist $r, s' \in Q$ and $r', s'_i \in F$ such that $q' = \gamma(q, \{r, s\})$, $r \xrightarrow{V_1} r'$ and $s \xrightarrow{W_1} s'$. Hence $V_1 \in L_A(r) - \{1\}$ and $W_1 \in L_A(s) - \{1\}$. Furthermore, by fork-acyclicity we have that $r, s \prec_A q$, so there exist series-rational expressions $e_r^+, e_s^+ \in \mathcal{T}_\Sigma$ such that $\llbracket e_r^+ \rrbracket = L_A(r) - \{1\}$ and $\llbracket e_s^+ \rrbracket = L_A(s) - \{1\}$. It follows that $V_1 \in \llbracket e_r^+ \rrbracket$ and $W_1 \in \llbracket e_s^+ \rrbracket$, and hence $V_1 \parallel W_1 \in \llbracket e_r^+ \parallel e_s^+ \rrbracket$. We conclude that $U = U_1 = V_1 \parallel W_1 \in \llbracket e_{qq'}^\emptyset \rrbracket \subseteq \llbracket e_{qq'}^{Q''} \rrbracket$.
- If $n > 1$. Then, clearly, Q'' is non-empty; let $q'' \in Q''$. Furthermore, let $I = \{0 \leq i \leq n : q_i = q''\}$. If $I = \emptyset$, then by the induction hypothesis we have that $U \in \llbracket e_{qq'}^{Q'' - \{q''\}} \rrbracket \subseteq \llbracket e_{qq'}^{Q''} \rrbracket$.

Otherwise, suppose that $I \neq \emptyset$, say $I = \{i_1, \dots, i_k\}$ with $i_1 < \dots < i_k$. Then, by the induction hypothesis, $U_{i_j+1} \cdots U_{i_{j+1}} \in \llbracket e_{q''q''}^{Q'' - \{q''\}} \rrbracket$ for all $1 \leq j < k$. Moreover, also by the induction hypothesis, $U_0 \cdots U_{i_1} \in \llbracket e_{qq''}^{Q'' - \{q''\}} \rrbracket$ and $U_{i_k+1} \cdots U_n \in \llbracket e_{q''q'}^{Q'' - \{q''\}} \rrbracket$. Hence,

$$U \in \llbracket e_{qq''}^{Q'' - \{q''\}} \cdot (e_{q''q''}^{Q'' - \{q''\}})^* \cdot e_{q''q'}^{Q'' - \{q''\}} \rrbracket \subseteq \llbracket e_{qq'}^{Q''} \rrbracket$$

We prove the implication from right to left also with induction on the size of Q'' . Suppose that $U \in \llbracket e_{qq'}^{Q''} \rrbracket$. If $Q'' = \emptyset$, then, considering the structure of $e_{qq'}^{Q''}$, there are two cases:

- If $U = a$ for some $a \in \Sigma$ such that $q' = \delta(q, a)$, then, clearly, $q \xrightarrow{U} q'$.
- If there exist $r, s \in Q$ and $V, W \in \text{Pom}_\Sigma^+$ such that $\{r, s\} \in \pi_A(q)$, $q' = \gamma(q, \{r, s\})$, $U = V \parallel W$, $V \in \llbracket e_r^+ \rrbracket$, and $W \in \llbracket e_s^+ \rrbracket$, then (by the premise) there exist $r', s' \in F$ such that $r \xrightarrow{V} r'$ and $s \xrightarrow{W} s'$. Hence, $q \xrightarrow{U} q'$.

Otherwise, suppose that Q'' is non-empty, and let $q'' \in Q''$. Then, again considering the structure of $e_{qq'}^{Q''}$, there are two cases:

- If $U \in \llbracket e_{qq''}^{Q'' - \{q''\}} \rrbracket$, then by the induction hypothesis $q \xrightarrow{U} q''$.
- If $U \in \llbracket e_{qq''}^{Q'' - \{q''\}} \cdot (e_{q''q''}^{Q'' - \{q''\}})^* \cdot e_{q''q'}^{Q'' - \{q''\}} \rrbracket$, then there exist $U', U_1, \dots, U_n$ ($n \geq 0$) and U'' such that $U = U' \cdot U_1 \cdots U_n \cdot U''$, $U' \in \llbracket e_{qq''}^{Q'' - \{q''\}} \rrbracket$, $U_i \in \llbracket e_{q''q''}^{Q'' - \{q''\}} \rrbracket$ for all $1 \leq i \leq n$, and $U'' \in \llbracket e_{q''q'}^{Q'' - \{q''\}} \rrbracket$. By the induction hypothesis, it follows that $q \xrightarrow{U'} q'' \xrightarrow{U_1} \cdots \xrightarrow{U_n} q'' \xrightarrow{U''} q'$, according to a series of paths that only visit states in $Q'' - \{q''\}$, with the intermediate states of these paths all q'' . We thus conclude that $q \xrightarrow{U} q'$ according to some path that only visits states in Q'' . $\square$

Using the auxiliary series-rational expressions $e_{qq'}^{Q''}$, we can now associate series-rational expressions $e_q, e_q^+ \in \mathcal{T}_\Sigma$ with every $q \in Q$, defining e_q^+ by $e_q^+ = \sum_{q' \in \rho_A(q) \cap F} e_{qq'}^{\rho_A(q)}$ and $e_q = e_q^+ + [q \in F]$. Note that $q \in \rho_A(q)$ and, by Lemma 3.9, for all $q' \in Q$ such that $q' \prec_A q''$ for some $q'' \in \rho_A(q)$ we have $q' \prec_A q$, and hence there exists, by induction, a series-rational expression $e_{q'} \in \mathcal{T}_\Sigma$ such that $\llbracket e_{q'} \rrbracket = L_A(q')$. So the expressions $e_{qq'}^{\rho_A(q)}$ are, indeed, defined in Definition 5.1.

Corollary 5.3. For every state $q \in Q$ we have $\llbracket e_q^+ \rrbracket = L_A(q) - \{1\}$ and $\llbracket e_q \rrbracket = L_A(q)$.

6 Discussion

Another automaton formalism for pomsets, *branching automata*, was proposed by Lodaya and Weil [13, 14]. Branching automata define the states where parallelism can start (*fork*) or end (*join*) in two relations; pomset automata condense this information in a single function. Lodaya and Weil also provided a translation of CKA expressions to branching automata, based on Thompson’s construction [16], which relies on the fact that their automata encode transitions non-deterministically, i.e., as *relations*. Our translation, in contrast, is based on Brzowski’s construction [2], and directly constructs transition *functions* from the expressions. Lastly, their translation of branching automata to CKA expressions is only sound for a *semantically* restricted class of automata, whereas our restriction is *syntactic*.

Jipsen and Moshier [7] provided an alternative formulation of the automata proposed by Lodaya and Weil, confusingly also called *branching automata*. Their method to encode parallelism in these branching automata is conceptually dual to pomset automata: branching automata distinguish based on the target states of traces to determine the join state, whereas pomset automata distinguish based on the origin states of traces. The translations of CKA expressions to branching automata and vice versa suffer from the same shortcomings as those by Lodaya and Weil. Specifically, the translation of expressions to automata similarly relied on non-determinism, and the translation from automata to expressions only worked for definition of language that was semantically restricted along the same lines as the semantic restriction put on (the earlier definition of) branching automata by Lodaya and Weil.

7 Further work

We plan to extend our results to CKA semantics in terms of downward-closed pomset languages, i.e., sets of pomsets that are closed under Gischer’s subsumption order [3]. Such an extension would correspond to adding the weak exchange law (which relates sequential and parallel compositions) to the axioms of CKA. We conjecture that our construction can be generalized to show the same correspondence between this semantics of series-rational expressions and pomset automata (i.e., without changes to the automaton model), just like Struth and Laurence suspect that the downward-closed semantics of series-parallel expressions can be captured by their non-downward closed semantics.

A classic result by Kozen [9] axiomatizes rational expressions using Kleene’s theorem [8] and the uniqueness of minimal finite automata. It would be interesting to see if the same technique can be used (based on pomset automata) to axiomatize series-rational expressions. Although an axiomatization of series-rational expressions was recently published [12], this proof does not rely on an automaton model. As such, it cannot be extended to axiomatize CKA with respect to the downward-closed semantics if the above conjecture holds.

Brzowski derivatives for classic rational expressions induce a coalgebra on rational expressions that corresponds to a finite automaton. We aim to study series-rational expressions coalgebraically. The first step would be to find the coalgebraic analogue of pomset automata such that language acceptance is characterized by the homomorphism into the final coalgebra. Ideally, such a view of pomset automata would give rise to a decision procedure for equivalence of series-rational expressions based on coalgebraic bisimulation-up-to [15].

Rational expressions can be extended with tests to reason about imperative programs equationally [11]. In the same vein, one can extend series-rational expressions with tests [6, 7] to reason about parallel imperative programs equationally. We are particularly interested in employing such an extension to extend the network specification language NetKAT [1] with primitives for concurrency so as to model and reason about concurrency within networks.

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A Proofs of auxiliary lemmas

Lemma 2.11. *Let $e, f \in \mathcal{T}_\Sigma$. If $e \simeq f$, then $\llbracket e \rrbracket = \llbracket f \rrbracket$. Also, $e \simeq 0$ if and only if $\llbracket e \rrbracket = \emptyset$.*

Proof. The proof of the first claim proceeds by induction on the construction of $\simeq$. In the base, there are nine cases to consider.

- If $e = f$, then the claim follows immediately.
- If $e = f + 0$, then $\llbracket e \rrbracket = \llbracket f \rrbracket \cup \llbracket 0 \rrbracket = \llbracket f \rrbracket \cup \emptyset = \llbracket f \rrbracket$.
- If $e = f + f$, then $\llbracket e \rrbracket = \llbracket f \rrbracket \cup \llbracket f \rrbracket = \llbracket f \rrbracket$.
- If $e = g_1 + g_2$ and $f = g_2 + g_1$, then $\llbracket e \rrbracket = \llbracket g_1 \rrbracket \cup \llbracket g_2 \rrbracket = \llbracket g_2 \rrbracket \cup \llbracket g_1 \rrbracket = \llbracket f \rrbracket$.
- If $e = g_1 + (g_2 + g_3)$ and $f = (g_1 + g_2) + g_3$, then $\llbracket e \rrbracket = \llbracket g_1 \rrbracket \cup (\llbracket g_2 \rrbracket \cup \llbracket g_3 \rrbracket) = (\llbracket g_1 \rrbracket \cup \llbracket g_2 \rrbracket) \cup \llbracket g_3 \rrbracket = \llbracket f \rrbracket$.
- If $e = 0 \cdot e_1$ and $f = 0$, then $\llbracket e \rrbracket = \llbracket 0 \rrbracket \cdot \llbracket e_1 \rrbracket = \emptyset = \llbracket f \rrbracket$.
- If $e = e_1 \cdot 0$ and $f = 0$, then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \cdot \llbracket 0 \rrbracket = \emptyset = \llbracket f \rrbracket$.

- If $e = 0 \parallel e_1$ and $f = 0$, then $\llbracket e \rrbracket = \llbracket 0 \rrbracket \parallel \llbracket e_1 \rrbracket = \emptyset = \llbracket f \rrbracket$.
- If $e = e_1 \parallel 0$ and $f = 0$, then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \parallel \llbracket 0 \rrbracket = \emptyset = \llbracket f \rrbracket$.

For the inductive step, there are six cases to consider.

- If $e \simeq f$ because $e \simeq g$ and $g \simeq f$, then $\llbracket e \rrbracket = \llbracket g \rrbracket$ and $\llbracket g \rrbracket = \llbracket f \rrbracket$ by induction, thus $\llbracket e \rrbracket = \llbracket f \rrbracket$.
- If $e \simeq f$ because $f \simeq e$, then $\llbracket f \rrbracket = \llbracket e \rrbracket$ by induction, and thus $\llbracket e \rrbracket = \llbracket f \rrbracket$.
- If $e \simeq f$ because $e = e_1 + e_2$ and $f = f_1 + f_2$ with $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then we know that $\llbracket e_1 \rrbracket = \llbracket f_1 \rrbracket$ and $\llbracket e_2 \rrbracket = \llbracket f_2 \rrbracket$ by induction. But then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \cup \llbracket e_2 \rrbracket = \llbracket f_1 \rrbracket \cup \llbracket f_2 \rrbracket = \llbracket f \rrbracket$.
- If $e \simeq f$ because $e = e_1 \cdot e_2$ and $f = f_1 \cdot f_2$ with $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then we know that $\llbracket e_1 \rrbracket = \llbracket f_1 \rrbracket$ and $\llbracket e_2 \rrbracket = \llbracket f_2 \rrbracket$ by induction. But then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \cdot \llbracket e_2 \rrbracket = \llbracket f_1 \rrbracket \cdot \llbracket f_2 \rrbracket = \llbracket f \rrbracket$.
- If $e \simeq f$ because $e = e_1 \parallel e_2$ and $f = f_1 \parallel f_2$ with $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then we know that $\llbracket e_1 \rrbracket = \llbracket f_1 \rrbracket$ and $\llbracket e_2 \rrbracket = \llbracket f_2 \rrbracket$ by induction. But then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \parallel \llbracket e_2 \rrbracket = \llbracket f_1 \rrbracket \parallel \llbracket f_2 \rrbracket = \llbracket f \rrbracket$.
- If $e \simeq f$ because $e = e_1^*$ and $f = f_1^*$ with $e_1 \simeq f_1$, then we know that $\llbracket e_1 \rrbracket = \llbracket f_1 \rrbracket$ by induction. But then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket^* = \llbracket f_1 \rrbracket^* = \llbracket f \rrbracket$.

We now prove the second claim. For the direction from left to right, note that $\llbracket e \rrbracket = \llbracket 0 \rrbracket = \emptyset$ by Lemma 2.11. The direction from right to left proceeds by induction on e . In the base, $e = 0$ and thus $e \simeq 0$ by reflexivity. For the inductive step, there are three cases to consider.

- If $e = e_1 + e_2$, then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \cup \llbracket e_2 \rrbracket = \emptyset$, therefore $\llbracket e_1 \rrbracket = \emptyset$ and $\llbracket e_2 \rrbracket = \emptyset$. But then, by induction, we have that $e_1 \simeq 0$ and $e_2 \simeq 0$. We then know that $e = e_1 + e_2 \simeq 0 + 0 \simeq 0$.
- If $e = e_1 \cdot e_2$, then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \cdot \llbracket e_2 \rrbracket = \emptyset$, therefore $\llbracket e_1 \rrbracket = \emptyset$ or $\llbracket e_2 \rrbracket = \emptyset$. But then, by induction, we have that $e_1 \simeq 0$ or $e_2 \simeq 0$. In either case, it follows that $e = e_1 \cdot e_2 \simeq 0$.
- If $e = e_1 \parallel e_2$, then $\llbracket e \rrbracket = \llbracket e_1 \rrbracket \parallel \llbracket e_2 \rrbracket = \emptyset$, therefore $\llbracket e_1 \rrbracket = \emptyset$ or $\llbracket e_2 \rrbracket = \emptyset$. But then, by induction, we have that $e_1 \simeq 0$ or $e_2 \simeq 0$. In either case, it follows that $e = e_1 \parallel e_2 \simeq 0$. $\square$

Lemma 3.7. *The reach of q contains $\{q' \in Q : \exists U \in \text{Pom}_\Sigma^+. q \xrightarrow{U}_A q'\} \cup \{\perp, q\}$. Moreover, if $\perp$ is the only state of A whose language is empty, this containment is an equality.*

Proof. We write $R_A(q) = \{q' \in Q : \exists U \in \text{Pom}_\Sigma^+. q \xrightarrow{U}_A q'\}$.

Suppose that $q' \in R_A(q) \cup \{q, \perp\}$. If $q' \in \{q, \perp\}$, then $q' \in \rho_A(q)$ by definition. Otherwise, if $q' \in R_A(q)$, then $q \xrightarrow{U}_A q'$ for some $U \in \text{Pom}_\Sigma^+$. We show, more generally, that if $q'' \xrightarrow{U}_\Sigma q'$ for some $q'' \in \rho_A(q)$, then $q' \in \rho_A(q)$, by induction on U . In the base, $U = a \in \Sigma$ and thus $q'' = \delta(q', a)$. But then $q' \in \rho_A(q)$ immediately. For the inductive step, there are two cases to consider.

- If $U = V \cdot W$ with V and W smaller than U , then there exists an $r \in Q$ such that $q'' \xrightarrow{V}_A r$ and $r \xrightarrow{W}_A q'$. By induction, $r \in \rho_A(q)$, and again by induction $q' \in \rho_A(q)$.
- If $U = V \parallel W$ with V and W smaller than U , then there exist $r, s \in Q$ and $r', s' \in F$ such that $r \xrightarrow{V}_A r'$ and $s \xrightarrow{W}_A s'$, and $q' = \gamma(q'', \{r, s\})$. Note that, in this case, $r', s' \neq \perp$. If $q' = \perp$, then $q' \in \rho_A(q)$ by definition. Otherwise, $\{r, s\} \in \pi_A(q'')$, and thus $q' \in \rho_A(q)$.

For the second part of the claim, suppose that $\perp$ is the only state of A with an empty language. Let $q' \in \rho_A(q)$. We prove that $q' \in R_A(q) \cup \{\perp, q\}$, by induction on the construction of $\rho_A(q)$. In the base, $q' = q$ or $q' = \perp$. In either case, we find that $q' \in R_A(q) \cup \{\perp, q\}$ immediately.

For the inductive step, there are two cases to consider. For both cases, first note that if $q'' \in R_A(q')$ and $q' \in R_A(q)$, then $q'' \in R_A(q)$.

- Suppose $q' = \delta(q'', a)$ for some $q'' \in \rho_A(q)$ and $a \in \Sigma$. By induction, we know that $q'' \in R_A(q) \cup \{q, \perp\}$. If $q'' = \perp$, then $q' = \perp$ and the claim follows. Otherwise, note that $q' \in R_A(q'')$ by $q'' \xrightarrow{a}_A q'$. If $q'' = q$, then $q' \in R_A(q)$ immediately. In the remaining case, $q'' \in R_A(q)$, and thus $q' \in R_A(q)$ by the observation above.
- Suppose $q' = \gamma(q'', \phi)$ for some $q'' \in \rho_A(q)$ and $\phi \in \pi_A(q'')$. By induction we know that $q'' \in R_A(q) \cup \{q, \perp\}$. Since $\phi \in \pi_A(q'')$, we know that $q'' \neq \perp$ — for otherwise $q' = \perp$ and thus $\phi \notin \pi_A(q'')$. Furthermore, if $\phi = \{r, s\}$ then $r, s \neq \perp$ and thus, since $\perp$ is the only state with an empty language, there exist $V, W \in \text{Pom}_\Sigma^+$ as well as $r', s' \in F$ such that $r \xrightarrow{V}_A r'$ and $s \xrightarrow{W}_A s'$; therefore, $q'' \xrightarrow{U \parallel V}_A q'$ and thus $q' \in R_A(q'')$. If $q'' = q$, then $q' \in R_A(q)$ immediately. Otherwise, $q'' \in R_A(q)$, and thus $q' \in R_A(q)$ by the observation above. $\square$

Lemma 4.4. *Let $e, f \in \mathcal{T}_\Sigma$ be such that $e \simeq f$. Then $e \in F_\Sigma$ if and only if $f \in F_\Sigma$.*

Proof. The proof proceeds by induction on the construction of $\simeq$. In the base, there are nine cases to consider.

- Suppose that $e = f$, then the claim holds immediately.
- Suppose that $e = f + 0$. If $e \in F_\Sigma$, then either $f \in F_\Sigma$ or $0 \in F_\Sigma$. Since the latter does not hold, $f \in F_\Sigma$ follows. Also, if $f \in F_\Sigma$, then $e = f + 0 \in F_\Sigma$ immediately.
- Suppose that $e = f + f$. If $f + f \in F_\Sigma$, then $f \in F_\Sigma$ immediately; if $f \in F_\Sigma$, then also $f + f \in F_\Sigma$.
- Suppose that $e = g_1 + g_2$ and $f = g_2 + g_1$. If $g_1 + g_2 \in F_\Sigma$, then $g_1 \in F_\Sigma$ or $g_2 \in F_\Sigma$; in either case, $g_2 + g_1 \in F_\Sigma$. The claim in the other direction is proved similarly.
- Suppose that $e = g_1 + (g_2 + g_3)$ and $f = (g_1 + g_2) + g_3$. If $e \in F_\Sigma$, then $g_1 \in F_\Sigma$ or $g_2 + g_3 \in F_\Sigma$, and thus one of g_1, g_2, g_3 must be in F_Σ . But then $g_1 + g_2$ or g_3 must be in F_Σ , and thus $f = (g_1 + g_2) + g_3 \in F_\Sigma$. The proof in the other direction is similar.
- Suppose that $e = 0 \cdot e_1$ and $f = 0$. Then $e, f \notin F_\Sigma$ and thus the claim holds immediately.
- Suppose that $e = e_1 \cdot 0$ and $f = 0$. Then $e, f \notin F_\Sigma$ and thus the claim holds immediately.
- Suppose that $e = e_1 \parallel 0$ and $f = 0$. Then $e, f \notin F_\Sigma$ and thus the claim holds immediately.
- Suppose that $e = 0 \parallel e_1$ and $f = 0$. Then $e, f \notin F_\Sigma$ and thus the claim holds immediately.

For the inductive step, there are six cases to consider.

- Suppose that $e \simeq f$ because $f \simeq e$; the claim (in both directions) then follows from the induction hypothesis by symmetry of mutual implication.
- Suppose that $e \simeq f$ because $e \simeq g$ and $g \simeq f$; the claim (in both directions) then follows from the induction hypothesis by transitivity of implication.
- Suppose that $e \simeq f$ because $e = e_1 + e_2$ and $f = f_1 + f_2$, with $e_1 \simeq f_1$ and $e_2 \simeq f_2$. If $e_1 + e_2 \in F_\Sigma$, either $e_1 \in F_\Sigma$ or $e_2 \in F_\Sigma$. But then (by induction) either $f_1 \in F_\Sigma$ or $f_2 \in F_\Sigma$, thus $f_1 + f_2 \in F_\Sigma$. The claim in the other direction is proved similarly.
- Suppose that $e \simeq f$ because $e = e_1 \cdot e_2$ and $f = f_1 \cdot f_2$, with $e_1 \simeq f_1$ and $e_2 \simeq f_2$. If $e \in F_\Sigma$, then $e_1, e_2 \in F_\Sigma$. But then (by induction) $f_1, f_2 \in F_\Sigma$, thus $f_1 \cdot f_2 \in F_\Sigma$. The claim in the other direction is proved similarly.
- Suppose that $e \simeq f$ because $e = e_1 \parallel e_2$ and $f = f_1 \parallel f_2$ with $e_1 \simeq f_1$ and $e_2 \simeq f_2$. If $e \in F_\Sigma$, then $e_1, e_2 \in F_\Sigma$. But then (by induction) $f_1, f_2 \in F_\Sigma$, thus $f_1 \parallel f_2 \in F_\Sigma$. The claim in the other direction is proved similarly.
- Suppose that $e \simeq f$ because $e = e_1^*$ and $f = f_1^*$ with $e_1 \simeq f_1$. Then $e, f \in F_\Sigma$, and so the claim holds immediately in both directions. $\square$

Lemma 4.5. *Let $e, f \in \mathcal{T}_\Sigma$ such that $e \simeq f$. If $a \in \Sigma$, then $\delta_\Sigma(e, a) \simeq \delta_\Sigma(f, a)$. Moreover, if $\phi = \{g, h\} \in \binom{\mathcal{T}_\Sigma}{2}$ with $g, h \neq 0$, then $\gamma_\Sigma(e, \phi) \simeq \gamma_\Sigma(f, \phi)$, and if $\psi \in \binom{\mathcal{T}_\Sigma}{2}$ with $\phi \simeq \psi$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e, \psi)$.*

Proof. The proof of the first claim proceeds by induction on the construction of $\simeq$. In the base, there are nine cases to consider.

- If $e = f$, then $\delta_\Sigma(e, a) = \delta_\Sigma(f, a) \simeq \delta_\Sigma(f, a)$.
- If $e = f + 0$, then $\delta_\Sigma(e, a) = \delta_\Sigma(f, a) + \delta_\Sigma(0, a) = \delta_\Sigma(f, a) + 0 \simeq \delta_\Sigma(f, a)$.
- If $e = f + f$, then $\delta_\Sigma(e, a) = \delta_\Sigma(f, a) + \delta_\Sigma(f, a) \simeq \delta_\Sigma(f, a)$.
- If $e = g_1 + g_2$ and $f = g_2 + g_1$, then

$$\delta_\Sigma(e, a) = \delta_\Sigma(g_1, a) + \delta_\Sigma(g_2, a) \simeq \delta_\Sigma(g_2, a) + \delta_\Sigma(g_1, a) = \delta_\Sigma(f, a)$$

- If $e = g_1 + (g_2 + g_3)$ and $f = (g_1 + g_2) + g_3$, then

$$\delta_\Sigma(e, a) = \delta_\Sigma(g_1, a) + (\delta_\Sigma(g_2, a) + \delta_\Sigma(g_3, a)) \simeq (\delta_\Sigma(g_1, a) + \delta_\Sigma(g_2, a)) + \delta_\Sigma(g_3, a) = \delta_\Sigma(f, a)$$

- If $e = 0 \cdot e_1$ and $f = 0$, then $\delta_\Sigma(e, a) = \delta_\Sigma(0, a) \cdot e_1 + 0 = 0 \cdot e_1 + 0 \simeq 0 = \delta_\Sigma(0, a)$.
- If $e = e_1 \cdot 0$ and $f = 0$, then $\delta_\Sigma(e, a) = \delta_\Sigma(e_1, a) \cdot 0 + e_1 \star \delta_\Sigma(0, a) \simeq 0 = \delta_\Sigma(0, a)$.
- If $e = e_1 \parallel 0$ and $f = 0$, then $\delta_\Sigma(e, a) = e_1 \star \delta_\Sigma(0, a) + 0 \simeq 0 = \delta_\Sigma(0, a)$.
- If $e = 0 \parallel e_1$ and $f = 0$, then $\delta_\Sigma(e, a) = 0 + e_2 \star \delta_\Sigma(0, a) \simeq 0 = \delta_\Sigma(0, a)$.

For the inductive step, there are six cases to consider.

- If $e \simeq f$ because $f \simeq e$, then by induction we know that $\delta_\Sigma(f, a) \simeq \delta_\Sigma(e, a)$, thus $\delta_\Sigma(e, a) \simeq \delta_\Sigma(f, a)$.
- If $e \simeq f$ because $e \simeq g$ and $g \simeq f$, then by induction we know that $\delta_\Sigma(e, a) \simeq \delta_\Sigma(g, a)$ and $\delta_\Sigma(g, a) \simeq \delta_\Sigma(f, a)$ thus $\delta_\Sigma(e, a) \simeq \delta_\Sigma(f, a)$.
- If $e \simeq f$ because $e = e_1 + e_2$ and $f = f_1 + f_2$ such that $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then by induction we know that $\delta_\Sigma(e_1, a) \simeq \delta_\Sigma(f_1, a)$ and $\delta_\Sigma(e_2, a) \simeq \delta_\Sigma(f_2, a)$, and thus

$$\delta_\Sigma(e, a) = \delta_\Sigma(e_1, a) + \delta_\Sigma(e_2, a) \simeq \delta_\Sigma(f_1, a) + \delta_\Sigma(f_2, a) = \delta_\Sigma(f, a)$$

- If $e \simeq f$ because $e = e_1 \cdot e_2$ and $f = f_1 \cdot f_2$ such that $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then by induction we know that $\delta_\Sigma(e_1, a) \simeq \delta_\Sigma(f_1, a)$ and $\delta_\Sigma(e_2, a) \simeq \delta_\Sigma(f_2, a)$, and thus

$$\delta_\Sigma(e, a) = \delta_\Sigma(e_1, a) \cdot e_2 + e_1 \star \delta_\Sigma(e_2, a) \simeq \delta_\Sigma(f_1, a) \cdot f_2 + f_1 \star \delta_\Sigma(f_2, a) = \delta_\Sigma(f, a)$$

where we also apply Lemma 4.4.

- If $e \simeq f$ because $e = e_1 \parallel e_2$ and $f = f_1 \parallel f_2$ such that $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then by induction we know that $\delta_\Sigma(e_1, a) \simeq \delta_\Sigma(f_1, a)$ and $\delta_\Sigma(e_2, a) \simeq \delta_\Sigma(f_2, a)$, and thus

$$\delta_\Sigma(e, a) = e_2 \star \delta_\Sigma(e_1, a) + e_1 \star \delta_\Sigma(e_2, a) \simeq f_2 \star \delta_\Sigma(f_1, a) + f_1 \star \delta_\Sigma(f_2, a) = \delta_\Sigma(f, a)$$

where we also apply Lemma 4.4.

- If $e \simeq f$ because $e = e_1^*$ and $f = f_1^*$ such that $e_1 \simeq f_1$, then by induction we know that $\delta_\Sigma(e_1, a) \simeq \delta_\Sigma(f_1, a)$, and thus $\delta_\Sigma(e, a) = \delta_\Sigma(e_1, a) \cdot e_1^* \simeq \delta_\Sigma(f_1, a) \cdot f_1^* = \delta_\Sigma(f, a)$.

The proof of the second claim also proceeds by induction on the construction of $\simeq$. In the base, there are nine cases to consider.

- If $e = f$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(f, \phi) \simeq \gamma_\Sigma(f, \phi)$.
- If $e = f + 0$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(f, \phi) + \gamma_\Sigma(0, \phi) = \gamma_\Sigma(f, \phi) + 0 \simeq \gamma_\Sigma(f, \phi)$.
- If $e = f + f$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(f, \phi) + \gamma_\Sigma(f, \phi) \simeq \gamma_\Sigma(f, \phi)$.
- If $e = g_1 + g_2$ and $f = g_2 + g_1$, then

$$\gamma_\Sigma(e, \phi) = \gamma_\Sigma(g_1, \phi) + \gamma_\Sigma(g_2, \phi) \simeq \gamma_\Sigma(g_2, \phi) + \gamma_\Sigma(g_1, \phi) = \gamma_\Sigma(f, \phi)$$

- If $e = g_1 + (g_2 + g_3)$ and $f = (g_1 + g_2) + g_3$, then

$$\gamma_\Sigma(e, \phi) = \gamma_\Sigma(g_1, \phi) + (\gamma_\Sigma(g_2, \phi) + \gamma_\Sigma(g_3, \phi)) \simeq (\gamma_\Sigma(g_1, \phi) + \gamma_\Sigma(g_2, \phi)) + \gamma_\Sigma(g_3, \phi) = \gamma_\Sigma(f, \phi)$$

- If $e = 0 \cdot e_1$ and $f = 0$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(0, \phi) \cdot e_1 + 0 = 0 \cdot e_1 + 0 \simeq 0 = \gamma_\Sigma(0, \phi)$.
- If $e = e_1 \cdot 0$ and $f = 0$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot 0 + e_1 \star \gamma_\Sigma(0, \phi) \simeq 0 = \gamma_\Sigma(0, \phi)$.
- If $e = 0 \parallel e_1$ and $f = 0$, then since $g, h \neq 0$, in particular $\phi \neq \{e_1, 0\}$. We then find that $\gamma_\Sigma(e, \phi) = 0 + e_1 \star \gamma_\Sigma(0, \phi) + 0 \simeq 0 = \gamma_\Sigma(0, \phi)$.

- If $e = e_1 \parallel 0$ and $f = 0$, then since $g, h \neq 0$, in particular $\phi \neq \{0, e_1\}$. We then find that $\gamma_\Sigma(e, \phi) = 0 + 0 + e_1 \star \gamma_\Sigma(0, \phi) \simeq 0 = \gamma_\Sigma(0, \phi)$.

For the inductive step, there are six cases to consider.

- If $e \simeq f$ because $f \simeq e$, then by induction we know that $\gamma_\Sigma(f, \phi) \simeq \gamma_\Sigma(e, \phi)$, thus $\gamma_\Sigma(e, \phi) \simeq \gamma_\Sigma(f, \phi)$.
- If $e \simeq f$ because $e \simeq g$ and $g \simeq f$, then by induction we know that $\gamma_\Sigma(e, \phi) \simeq \gamma_\Sigma(g, \phi)$ and $\gamma_\Sigma(g, \phi) \simeq \gamma_\Sigma(f, \phi)$, thus $\gamma_\Sigma(e, \phi) \simeq \gamma_\Sigma(f, \phi)$.
- If $e \simeq f$ because $e = e_1 + e_2$ and $f = f_1 + f_2$ such that $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then by induction we know that $\gamma_\Sigma(e_1, \phi) \simeq \gamma_\Sigma(f_1, \phi)$ and $\gamma_\Sigma(e_2, \phi) \simeq \gamma_\Sigma(f_2, \phi)$, and thus

$$\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) + \gamma_\Sigma(e_2, \phi) \simeq \gamma_\Sigma(f_1, \phi) + \gamma_\Sigma(f_2, \phi) = \gamma_\Sigma(f, \phi)$$

- If $e \simeq f$ because $e = e_1 \cdot e_2$ and $f = f_1 \cdot f_2$ such that $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then by induction we know that $\gamma_\Sigma(e_1, \phi) \simeq \gamma_\Sigma(f_1, \phi)$ and $\gamma_\Sigma(e_2, \phi) \simeq \gamma_\Sigma(f_2, \phi)$, and thus

$$\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot e_2 + e_1 \star \gamma_\Sigma(e_2, \phi) \simeq \gamma_\Sigma(f_1, \phi) \cdot f_2 + f_1 \star \gamma_\Sigma(f_2, \phi) = \gamma_\Sigma(f, \phi)$$

where we also apply Lemma 4.4.

- If $e \simeq f$ because $e = e_1 \parallel e_2$ and $f = f_1 \parallel f_2$ such that $e_1 \simeq f_1$ and $e_2 \simeq f_2$, then by induction we know that $\gamma_\Sigma(e_1, \phi) \simeq \gamma_\Sigma(f_1, \phi)$ and $\gamma_\Sigma(e_2, \phi) \simeq \gamma_\Sigma(f_2, \phi)$. Furthermore, $\phi \simeq \{e_1, e_2\}$ if and only if $\phi \simeq \{f_1, f_2\}$, and thus

$$\begin{aligned} \gamma_\Sigma(e, \phi) &= [\{e_1, e_2\} \simeq \phi] + e_2 \star \gamma_\Sigma(e_1, \phi) + e_1 \star \gamma_\Sigma(e_2, \phi) \\ &\simeq [\{f_1, f_2\} \simeq \phi] + f_2 \star \gamma_\Sigma(f_1, \phi) + f_1 \star \gamma_\Sigma(f_2, \phi) \\ &= \gamma_\Sigma(f, \phi) \end{aligned}$$

where we also apply Lemma 4.4.

- If $e \simeq f$ because $e = e_1^*$ and $f = f_1^*$ such that $e_1 \simeq f_1$, then by induction we know that $\gamma_\Sigma(e_1, \phi) \simeq \gamma_\Sigma(f_1, \phi)$, and thus $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot e_1^* \simeq \gamma_\Sigma(f_1, \phi) \cdot f_1^* = \gamma_\Sigma(f, \phi)$.

The last claim is shown by induction on e . In the base, $e = 0$, $e = 1$ or $e = a$, and thus $\gamma_\Sigma(e, \phi) = 0 = \gamma_\Sigma(e, \psi)$. For the inductive step, there are four cases to consider.

- If $e = e_1 + e_2$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) + \gamma_\Sigma(e_2, \phi) = \gamma_\Sigma(e_1, \psi) + \gamma_\Sigma(e_2, \psi) = \gamma_\Sigma(e, \psi)$.
- If $e = e_1 \cdot e_2$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot e_2 + e_1 \star \gamma_\Sigma(e_2, \phi) = \gamma_\Sigma(e_1, \psi) \cdot e_2 + e_1 \star \gamma_\Sigma(e_2, \psi) = \gamma_\Sigma(e, \psi)$.
- If $e = e_1 \parallel e_2$, then $\gamma_\Sigma(e, \phi) = [\{e_1, e_2\} \simeq \phi] + e_2 \star \gamma_\Sigma(e_1, \phi) + e_1 \star \gamma_\Sigma(e_2, \phi) = [\{e_1, e_2\} \simeq \psi] + e_2 \star \gamma_\Sigma(e_1, \psi) + e_1 \star \gamma_\Sigma(e_2, \psi)$.
- If $e = e_1^*$, then $\gamma_\Sigma(e, \phi) = \gamma_\Sigma(e_1, \phi) \cdot e_1^* = \gamma_\Sigma(e_1, \psi) \cdot e_1^* = \gamma_\Sigma(e, \psi)$. □

Lemma 4.21. *Let $e, f \in \mathcal{T}_\Sigma$ and $U \in \mathbf{Pom}_\Sigma^+$. If $e \xrightarrow{U}_\Sigma f$, then $[e] \xrightarrow{U}_\simeq [f]$. If $[e] \xrightarrow{U}_\simeq [f]$, then there exists an $f' \in \mathcal{T}_\Sigma$ with $f \simeq f'$ and $e \xrightarrow{U}_\Sigma f'$.*

Proof. The proof for the first claim proceeds by induction on U . In the base, $U = a \in \Sigma$. If $e \xrightarrow{U}_\Sigma f$, then $f = \delta_\Sigma(e, a)$. But then, since $\delta_\simeq([e], a) = [f]$ by definition of $\delta_\simeq$, we find that $[e] \xrightarrow{U}_\simeq [f]$.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$, with V and W smaller than U , then there exists a $g \in \mathcal{T}_\Sigma$ such that $e \xrightarrow{V}_\Sigma g$ and $g \xrightarrow{W}_\Sigma f$. By induction we find that $[e] \xrightarrow{V}_\simeq [g]$ and $[g] \xrightarrow{W}_\simeq [f]$, and thus $[e] \xrightarrow{U}_\simeq [f]$.
- If $U = V \parallel W$, with V and W smaller than U , then there exist $g, h \in \mathcal{T}_\Sigma$ and $g', h' \in F_\Sigma$ such that $g \xrightarrow{V}_\Sigma g'$ and $h \xrightarrow{W}_\Sigma h'$, and $f = \gamma_\Sigma(e, \{g, h\})$. Note that, by Lemma 4.7, $g, h \neq 0$. By induction, we find $[g] \xrightarrow{V}_\simeq [g']$ and $[h] \xrightarrow{W}_\simeq [h']$, with $[g'], [h'] \in F_\simeq$. Since $\gamma_\simeq([e], \{[g], [h]\}) = [\gamma_\Sigma(e, \{g, h\})] = [f]$, we find that $[e] \xrightarrow{U}_\simeq [f]$.

The proof of the second claim also proceeds by induction on U . In the base, $U = a \in \Sigma$. If $[e] \xrightarrow{U}_\Sigma [f]$, then $[f] = \delta_\simeq([e], a)$. We choose $f' = \delta_\Sigma(e, a)$ to find that $e \xrightarrow{U}_\Sigma f'$ and $f \simeq f'$ by Lemma 4.5.

For the inductive step, there are two cases to consider.

- If $U = V \cdot W$, with V and W smaller than U , then there exists a $g \in \mathcal{T}_\Sigma$ such that $[e] \xrightarrow{V} \simeq [g]$ and $[g] \xrightarrow{W} \simeq [f]$. By induction, we find $g', f'' \in \mathcal{T}_\Sigma$ such that $e \xrightarrow{V} g'$ and $g \simeq g'$ as well as $g \xrightarrow{W} f''$ and $f \simeq f''$. By Lemma 4.6, we find $f' \in \mathcal{T}_\Sigma$ such that $g' \xrightarrow{W} f'$ and $f'' \simeq f$. In total, we have that $e \xrightarrow{U} f'$ with $f \simeq f'$.
- If $U = V \parallel W$, with V and W smaller than U , then there exist $[g], [h] \in \mathcal{Q}_\Sigma$ and $[g'], [h'] \in F_\simeq$ such that $[g] \xrightarrow{V} \simeq [g']$ and $[h] \xrightarrow{W} \simeq [h']$, and $[f] = \gamma_\simeq([e], \{[g], [h]\})$. By induction, we find that $h'', g'' \in \mathcal{T}_\Sigma$ such that $g \xrightarrow{V} g''$ and $h \xrightarrow{W} h''$, with $g' \simeq g''$ and $h' \simeq h''$, with $g'', h'' \in F_\Sigma$ by Lemma 4.4. But then $g, h \not\simeq 0$ by Lemma 4.7. We can choose $f' = \gamma_\Sigma(e, \{g, h\})$ to find that that $e \xrightarrow{U} f'$ and $f \simeq f'$ by Lemma 4.5. $\square$

Lemma 4.25. *Let $e, e' \in \mathcal{T}_\Sigma$ be such that $e \simeq e'$. Then $\langle e \rangle = \langle e' \rangle$.*

Proof. First, if $e \simeq 0 \simeq e'$, then $\langle e \rangle = 0 = \langle e' \rangle$ by definition. The proof for the remaining cases proceeds by induction on the construction of $\simeq$. In the base, there are nine cases to consider.

- If $e = e'$, then the claim holds immediately.
- If $e = e' + 0$, then $\langle e \rangle = \max(\langle e' \rangle, \langle 0 \rangle) = \langle e' \rangle$.
- If $e = e' + e'$, then $\langle e \rangle = \max(\langle e' \rangle, \langle e' \rangle) = \langle e' \rangle$.
- If $e = e_1 + e_2$ and $e' = e_2 + e_1$, then $\langle e \rangle = \max(\langle e_1 \rangle, \langle e_2 \rangle) = \max(\langle e_2 \rangle, \langle e_1 \rangle) = \langle e' \rangle$.
- If $e = e_1 + (e_2 + e_3)$ and $e' = (e_1 + e_2) + e_3$, then $\langle e \rangle = \max(\langle e_1 \rangle, \langle e_2 \rangle, \langle e_3 \rangle) = \langle e' \rangle$.

For the inductive step, there are six cases to consider.

- If $e \simeq e'$ because $e' \simeq e$, then $\langle e' \rangle = \langle e \rangle$ by induction and so the claim follows.
- If $e \simeq e'$ because $e \simeq e'' \simeq e'$, then $\langle e \rangle = \langle e'' \rangle = \langle e' \rangle$ by induction and so $\langle e \rangle = \langle e' \rangle$.
- If $e \simeq e'$ because $e = e_1 + e_2$ and $e' = e'_1 + e'_2$ with $e_1 \simeq e'_1$ and $e_2 \simeq e'_2$, then $\langle e_1 \rangle = \langle e'_1 \rangle$ and $\langle e_2 \rangle = \langle e'_2 \rangle$ by induction, thus $\langle e \rangle = \max(\langle e_1 \rangle, \langle e_2 \rangle) = \max(\langle e'_1 \rangle, \langle e'_2 \rangle) = \langle e' \rangle$.
- If $e \simeq e'$ because $e = e_1 \cdot e_2$ and $e' = e'_1 \cdot e'_2$ with $e_1 \simeq e'_1$ and $e_2 \simeq e'_2$, then $\langle e_1 \rangle = \langle e'_1 \rangle$ and $\langle e_2 \rangle = \langle e'_2 \rangle$ by induction, thus $\langle e \rangle = \max(\langle e_1 \rangle, \langle e_2 \rangle) = \max(\langle e'_1 \rangle, \langle e'_2 \rangle) = \langle e' \rangle$.
- If $e \simeq e'$ because $e = e_1 \parallel e_2$ and $e' = e'_1 \parallel e'_2$ with $e_1 \simeq e'_1$ and $e_2 \simeq e'_2$, then $\langle e_1 \rangle = \langle e'_1 \rangle$ and $\langle e_2 \rangle = \langle e'_2 \rangle$ by induction, thus $\langle e \rangle = \max(\langle e_1 \rangle, \langle e_2 \rangle) + 1 = \max(\langle e'_1 \rangle, \langle e'_2 \rangle) + 1 = \langle e' \rangle$.
- If $e \simeq e'$ because $e = e_1^*$ and $e' = e_1'^*$, then by induction $\langle e \rangle = \langle e_1 \rangle = \langle e_1' \rangle = \langle e' \rangle$. $\square$